

On Quantum Algorithms for Weyl Systems, Modular Subset Sum Problem Type Problems, and an Architectural Complexity Separation with an Application of Architectural Complexity Separation

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Written 26th August 2025*

Revised 14th August 2026*

Draft version

(Dates of server timestamps)*

Abstract

The verification of high-dimensional modular systems involving irrational constants is a challenge in computational number theory typically requiring $O(2^n)$ classical operations. Here, we introduce the quantum algorithmic framework "the Wobbly Quantum Fourier Transform (WQFT)" that does this verification to polynomial time. By exploiting the isomorphism between the tensor-product structure of the QFT and the power-set expansion of irrational modular equations, we demonstrate that a quantum system can sense global phase stability. We provide a derivation of the convergence bounds and describe an algorithm to discern the constructive interference of solutions and the Weyl-distributed noise of non-solutions.

We give a proof, in the appendix, that shows the two types of problems our quantum algorithm solves can be reduced to the known generic subset sum type problems (known NP-complete problems). While each individual constraint is NP-complete, the verification of the global Weyl system requires the simultaneous, consistent satisfaction of $SO(2^n)$ such instances. Consequently, the classical verification requires $O(2^n)$ operations, placing the global problem outside of NP and within EXP.

Algorithm 2 introduces a new type of amplitude amplification. All our algorithms are in the ideal gate model of quantum computing. We claim, in our ACS framework, that our quantum algorithm is faster than some purely classical way to solve the Weyl system of equations classically. Also the purpose of this paper is to present a conceptually different faster quantum way than compared to the purely classical brute force type algorithm presented here. Our above algorithm is based on conjecture 1 which is related to a proof of separation of MA and QCMA and the above proof is also related to the "promise gap" described at the top of page 16. While this research offers a foundation that may eventually inform a formal, traditional complexity-theoretic proof, we introduce a new framework: Architectural Complexity Separation (ACS). This framework sits at the intersection of empirical algorithmic benchmarking and traditional existential complexity proofs, providing a robust, instance-based model of separation that bridges the gap between pure theory and the practical realities of quantum verification.

We demonstrate a verifiable (Architectural Complexity) separation between the complexity classes MA and QCMA. We demonstrate that the Weyl system

problem is in BQP (via the WQFT algorithm) and in QCMA (via the ability to verify a classical witness using quantum interference). Crucially, the classical verification of this problem requires $O(2^n)$ operations, placing it outside of NP and within EXP. This provides a concrete instance-based architectural separation: a problem that is efficiently solvable/verifiable in QCMA but intractable for classical deterministic or probabilistic architectures.

Section i: On Foundational Priority Foundational Priority Declaration for the Architectural Complexity Separation Paradigm

This paper establishes the foundational architecture, algorithmic primitives, and complexity-theoretic framework underlying Architectural Complexity Separation (ACS) for high-dimensional Weyl modular systems. It introduces and formalizes the complete structural, algebraic, and quantum-computational mechanisms required to verify exponentially large modular systems, systems whose classical verification intrinsically resides outside NP and within EXP, using only polynomial quantum resources.

Within this foundational declaration, we introduce and unify the core constructs of the ACS paradigm, including but not limited to: Wobbly Quantum Fourier Transform (WQFT), latent QFT, Kronecker-isomorphic phase accumulation, global phase-manifold verification, latent QFT emergence, the dual Weyl System Oracle Types (WSOT1 and WSOT2), tensor-product resonance, irrational-to-rational phase alignment, and the structural trap imposed by non-separable modular constraints. These mechanisms collectively define the architectural substrate through which quantum systems achieve polynomial-time verification of problems that require $O(2^n)$ extensional checks on classical machines.

This paper further establishes the first un-relativized, white-box architectural separation between QCMA and MA in the ACS framework. Unlike some other oracle-based separations, which rely on relativized black-box constructions, our separation is achieved through a fully explicit verification algorithm, a white-box quantum procedure whose internal structure is completely specified and whose correctness is grounded in the tensor-product geometry of the Weyl system. The ACS framework therefore provides a new model for demonstrating separations: not by oracle relativization, but by architectural obstruction, where classical machines are structurally trapped by non-separable modular noise while quantum machines exploit global phase symmetries.

This foundational claim explicitly encompasses the full development of the ACS framework, the MA-QCMA architectural separation, and the quantum verification primitives introduced in Algorithms 1 and 2. Algorithm 1 establishes the Null-Resonance Phase Masking Primitive, demonstrating how global modular error cancellation restores periodic symmetry detectable via QFT. Algorithm 2 establishes the Latent QFT Gradient Primitive, showing how local irrational rotations can be tuned to synthesize a rational Fourier manifold across the entire 2^n -dimensional Hilbert space. Together, these primitives form the architectural backbone of the ACS paradigm.

The framework further includes the heuristic reductions to modular subset-sum type problems, the extensional verification bottleneck created by non-linear modulo-1 collapse, the Kronecker-tensor structural isomorphism, and the explicit demonstration that the Weyl verification problem, despite being reducible to NP-complete components, requires exponential classical verification and is therefore outside NP and within EXP, while simultaneously residing in BQP and QCMA under quantum verification.

This paper defines the foundational structure of the ACS paradigm and its quantum primitives. Future works may refine proofs, extend special cases, or introduce additional variants of the WQFT architecture, but such developments are derivative elaborations of the unified framework established here. The ACS model, the WQFT/latent QFT primitives, and the Weyl-system verification architecture together constitute the complete foundational basis for quantum-intrinsic verification of high-dimensional modular systems and for the architectural separation between classical and quantum complexity classes.

Section 0: Architectural Complexity Separation (ACS)

0.1 Framework Definition

This paper assumes the widely believed to be true conjecture that some quantum computer/system cannot be simulated on a purely classically computer, Traditional complexity-theoretic separations (e.g., MA vs. $QCMA$) often rely on existential proofs within idealized oracles. While theoretically rigorous, these abstractions often fail to capture the architectural constraints that define quantum advantage in real-world systems. We propose Architectural Complexity Separation (ACS) as an empirical framework to characterize the gap between computational models.

A language L exhibits a ACS if there exists a verifiable problem instance $I \in L$ constructed such that:

1. **The Structural Trap:** For a classical model $\mathcal{M}_C$, any verification algorithm for instance I is subject to an $O(2^n)$ complexity lower bound, imposed by non-local, non-separable interdependencies.
2. **The Asymmetric Key:** For a quantum model $\mathcal{M}_Q$, instance I possesses an algebraic symmetry "such as tensor product resonance "that permits polynomial-time verification.
3. **The Falsifiable Boundary:** The structural trap is explicitly defined; the separation stands as a verifiable benchmark until a classical algorithm is demonstrated to solve the instance I in polynomial time.

0.2 The Predictability Paradox

This work demonstrates that the verification of high-dimensional Weyl modular systems satisfies the ACS criteria. We utilize Kronecker phase accumulation ($\Phi_x = \sum x_j \chi_j$) as our Structural Trap, forcing classical verifiers into exhaustive search spaces. Conversely, we utilize the Wobbly Quantum Fourier Transform (WQFT) as our Asymmetric Key, treating the collective noise not as stochastic interference, but as a measurable global phase manifold.

This resolves the apparent paradox of "predicting an unpredictable noisy system." We posit that unpredictability is not an intrinsic property of the Weyl system, but a consequence of the verifier's architectural limitations. To a classical verifier constrained by local modularity, the system is indeed unpredictable. However, for a quantum observer, the randomness is merely a phase-shifted manifold that is structurally stable. The "oxymoron" of our result is resolved by recognizing that the system's predictability is observer-dependent: our architecture allows the quantum machine to bypass the noise floor that traps the classical machine.

0.3 Empirical Basis and Classical Baseline

The validity of this ACS is substantiated by the historical intractability of high-dimensional Weyl modular systems. Despite the efficacy of standard classical reduction algorithms, such as LLL, these tools fail to verify these systems due to two structural barriers:

- **Vector Stretching:** The linear bounds of the modular parameters in our system 'stretch' the solution vector, increasing the complexity beyond the convergence bounds of greedy lattice-reduction heuristics.
- **Local vs. Global Nature:** LLL is a greedy approach designed to find a local short vector. Weyl modular systems are inherently degenerate, requiring the global phase stability that only the WQFT can extract.

The persistent inability of classical approaches to bypass the noise floor created by Kronecker phase accumulation provides empirical weight to the existence of a CCS boundary. Until a classical algorithm is formulated that resolves these systems in polynomial time, this construction serves as an empirical verification of the structural separation between *MA* and *QCMA*.

The history of quantum complexity is dominated by the discovery of natural problems that admit quantum advantage, such as factoring via Shor's algorithm. These results, while groundbreaking, rely on inherent mathematical properties (e.g., period-finding). In contrast, we propose a shift toward **Architectural Complexity Separation (ACS)**. Instead of searching for natural-born quantum-hard problems, ACS defines a framework for *engineering* verifiable instances where a classical verifier is structurally blocked by non-local noise. While Shor's algorithm exposes a natural weakness in classical complexity, our work defines a new class of complexity barrier, the noisy Weyl system, which persists

as a challenge until a classical architecture can bypass the global phase manifold. This is not merely an algorithm; it is the establishment of a benchmark for structural verification.

In contrast to prior oracle based separations, we give a fully unrelativized separation of QCMA from MA within the ACS framework, using a natural white-box verification problem whose classical hardness is supported by a rigorous NP-complete lower bound.

On the Function $\mathbb{I}(i, j)$

The indicator type function $\mathbb{I}(i, j)$ extracts the j -th bit of the index i . The function $\mathbb{I}(i, j)$ is shorthand for the function $\mathbb{I}(\text{Permute}(i), j)$. In other words, the function $\mathbb{I}(i, j)$ contains the Permute function but instead of writing $\mathbb{I}(\text{Permute}(i), j)$ we write $\mathbb{I}(i, j)$ so the function is both an indicator and Permute function.

The system of equations solved by algorithm 1 and 2 is defined over a general 1-to-1 permutation (bijection), the Permute function.

$$\text{Permute} : \{0, 1\}^k \rightarrow \{0, 1\}^k$$

While there are many choices to choose for $\text{Permute}(X)$, e.g. one choice can use bit rotation, before Algorithm 1 or Algorithm 2 are used, for this paper we use the following choice for $\text{Permute}(X)$:

$$\text{Permute}(X) = (\text{Reverse}(X) \oplus C)$$

Where $\text{Reverse}(X)$ is a function that reverses the binary string. C is a constant chosen before algorithm 1 or algorithm 2 is run, for our paper we choose $C=3$, but C can be any other value less which does not exceed the maximal binary length of i .

The example choice of $(\text{Reverse}(X) \oplus C)$ is chosen to simplify the proof in the appendix that the new verifiable oracle works in polynomial space and polynomial time but a more complicated example for $\text{Permute}(X)$ can easily be chosen. For example we could construct a choice for $\text{Permute}(X)$ based on using a primitive root, for a prime one greater than a power of 2 (e.g. a Fermat Prime) to some chosen base, the base then raised to the power of X and then XOR this with X .

Introduction - Setting the Scene Of How the Algorithms Work

Algorithm 1 works by injecting the Weyl equation parameters directly into the phase exponents of a global diagonal unitary operator. The oracle applies a composite transformation $U = B \cdot A$, where the left-hand matrix uses parallel single-qubit rotations to populate the exponential subset sums, while matrix B encodes the target offsets. This maps the modular system's accumulated

error, $\delta_x = (\sum n_j \gamma_j \mathbb{I}(x, j) - \alpha_x)$, straight into the global wavefunction. When a true solution is evaluated, $\delta_x \rightarrow 0 \pmod{1}$, making the phase mask completely transparent and preserving the register's underlying periodic symmetry. In this exact limit, an ideal unperturbed state is reached, allowing a downstream inverse QFT to gather the synchronized phases into a single resonant frequency peak.

Algorithm 2 works by injecting the Weyl equations directly into the phase exponents of parallel, single-qubit gates, establishing a uniform superposition $(|+\rangle)^{\otimes n}$ as its unique physical anchor. By computing rotation angles locally as $\theta_j = (n_j \gamma_j - \alpha_j) \pmod{2\pi}$, the oracle leverages Kronecker tensor geometry to automatically multiply the individual phase factors across the register. When a true solution is found, the irrational constants balance perfectly, causing this phase anchor to "snap" onto a clean rational grid ($\theta_j \rightarrow \frac{2\pi}{2^j}$). In this exact limit, the unentangled gate array converges functionally to a dense, non-zero QFT state, engineering a global linear phase gradient across the exponential manifold that an external detector decodes.

The Weyl verifications, for each of both types of WSOT(Weyl System of Type), is a satisfiability decision problem within a critical constraint density regime. The system, in each WSOT, is well-posed, with an injective phase mapping for non-degenerate irrational parameters. Multiple valid solution vectors define a complex manifold, consistent with hard decision problems. Crucially, the Weyl verification problem, in each WSOT, is computationally distinct from traditional NP-complete classes. Unlike NP-complete problems permitting polynomial-time verification, Weyl systems exhibit an Extensional Verification Bottleneck; global phase stability cannot be validated through local queries; In each WSOT, it requires checking the entire power set. This necessitates an Architectural Complexity Separation (ACS). While classical architectures are trapped in an inherently exponential $O(2^n)$ verification process due to this non-separable global constraint, the quantum WQFT/latent QFT framework efficiently verifies the system in polynomial time by operating on the global state manifold rather than the power set, effectively bypassing the specific bottleneck that restricts all classical computational verification methods.

We refer to the above system of equation in the problem above as a Weyl system in problem 1 and 2 by Algorithm 1 and 2 respectively. The both above problems reduces (see appendix I) to a modular subset sum (type) problem which is problem known to be NP-complete(so has no polynomial time algorithm) so the Weyl system(used in problems 1 and 2) is at least as difficult as the sub set sum problem, as shown in proof appendix E, this reduction proof does not need use the wrap-around modulo 2π property of the Weyl System but such a proof using the wrap-around modulo 2π property of the Weyl System is discussed in appendix E.

Suggestion for Choosing Values for γ_j, α_j For Numerical Simulation of Algorithms 1 and 2

We may choose γ_j, α_j constants to be recognisable algebraic independent numbers (some integer root - to be chosen- of rational number for example.) for the

purposes of numerical simulation.

Statement of Problem Solved by Algorithm 1

For Algorithm 1, we define a system of $M = 2^n$ modular equations where the objective is to verify if a candidate vector $n \in \{0, 1\}^n$ satisfies the null-congruence $\delta_i \approx 0 \pmod{1}$.

n_j must satisfy the linear scaling constraints :
the coefficients n_j , such that $n_j \in (2(j-1) \cdot u, 2j \cdot u]$ for a sufficiently large base $u \in \mathbb{Z}^+$. So the problem also requires to find a suitable value of u along with the all the n_j .

This problem (that is solved by algorithm 1) requires **that all 2^n global modular equations simultaneously satisfy** $\max |\delta_x| < \epsilon$, (or $\max |\delta_x| < \frac{\epsilon_{p1}}{2^n}$ in appendix G $\frac{\epsilon_{p1}}{2^n} = \epsilon$) **with** $0 \leq x < (-1 + (2^h))$, **for some integer h** , the quantum computer evaluates this boundary implicitly via the QFT interference using phase errors of the QFT. Note ϵ_{p1} can be set for each individual n and $\epsilon_{p1} = O(\frac{1}{2^n} \log n)$ at most for growth for Appendix G to apply (in polynomial time and polynomial space). ϵ_{p1} mean episilon for problem 1.

See appendix D for an explanation of how $\max |\delta_x| < \epsilon$, with $0 \leq x < (-1 + (2^h))$ is related to the QFT.

we refer to the above system of equation in the problem above as a Weyl system. The above problem reduces (see appendix I) to a modular subset sum (type) problem which is problem known to be NP-complete.

1. The Core Operators

The verification is performed by the unitary operator $U = B \cdot A$, where:

- **Operator A (LHS Matrix):**

$$A = \bigotimes_{j=1}^n R_j, \quad R_j = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i n_j \gamma'_j} \end{pmatrix}$$

Via the Kronecker product, the diagonal entries are:

$$(A)_{ii} = \exp \left(2\pi i \sum_{j=1}^n n_j \gamma'_j \mathbb{I}(i, j) \right)$$

This “embodies” the Left-Hand Side subset sums for all 2^n equations simultaneously.

- **Operator B (RHS/Target Matrix):**

$$B = \text{diag}(e^{-2\pi i\alpha_0}, e^{-2\pi i\alpha_1}, \dots, e^{-2\pi i\alpha_{M-1}})$$

This encodes the target offsets α_i (the Right-Hand Side).

2. The QFT Generalization Equation

The algorithm applies U to a periodic anchor state $|\psi_p\rangle$. The probability amplitude $A(y)$ for measuring frequency y is a generalization of the standard QFT, where the modular error δ_i is embedded "beside" the Fourier factor:

2. Interaction: The "Wobbly" Matrix Assembly

The interaction between the irrational subset sums and the α constants happens inside the tensor factors before the global matrix is even formed. The algorithm constructs the operator $U(n)$ by "subtracting" the alpha phases directly from the irrational phases at the local qubit level:

$$U(n) = \bigotimes_{j=1}^n \frac{1}{\sqrt{2}} \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & e^{i(n_j\gamma'_j - \alpha_j)} \end{pmatrix}}_{R_j(\text{Adjusted Factor})}$$

Description of Problem Solved by Algorithm 2

In the Latent QFT framework of Algorithm 2, the α constants are not just random offsets; they are the Phase Aligners that allow the Kronecker product to "snap" into the QFT structure. n_j must satisfy the linear scaling constraints: the coefficients n_j , such that $n_j \in (2j-1)*u, 2j*u]$ for a sufficiently large base $u \in \mathbb{Z}^+$. So the problem also requires to find a suitable value of u along with the all the n_j .

1. The Alpha Matrix (B)

Because Algorithm 2 relies on the n -fold tensor product to generate the 2^n equations, the α values must also follow a specific structure to avoid breaking the tensor factoring. We define the Target Phase Matrix B as a diagonal matrix:

$$B = \text{diag}(e^{-i\alpha_0}, e^{-i\alpha_1}, \dots, e^{-i\alpha_{M-1}})$$

To maintain the "Latent QFT" property, this matrix B maybe Kronecker product of n local alpha-shifts:

$$B = \bigotimes_{j=1}^n \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\alpha_j} \end{pmatrix}$$

This ensures that the target side of our 2^n equations "involving α_i " are actually combinations of n base α_j constants. Note that the matrix need not take the form above of being decomposable into smaller matrices; B can take exponential time and exponential space to compute but the matrix B does not have an impact on a main result of this paper which is our verification oracle will still work in polynomial time and polynomial space. Also we give a version of the problem with noise and where B the above Kronecker product factorization in appendix F.

Statement of Problem Solved by Algorithm 2

This Problem (solved by algorithm 2) requires the maximum local phase deviation across the register to satisfy $\max_j |\theta_j - \frac{\pi}{2^z}| < \epsilon$, $0 \leq j < (-1 + 2^z)$, let $\epsilon_{p2} = \epsilon$, ϵ_{p2} grows as the functioned mentioned below, for some integer z , the quantum register naturally maps this constraint to its physical geometry, using the inverse QFT to automatically collapse the resonance peak if any single coordinate breaches the threshold where $\theta_j = (n_j \gamma_j - \alpha_j)$ and note that n_j behaves like a pseudo random number and not a function of j but is labelled by j . See appendix D for an explanation of how $\max_j |\theta_j - \frac{\pi}{2^z}| < \epsilon$, $0 \leq j < (-1 + 2^z)$ is related to the QFT. We refer to the system of equations in the above problem as a Weyl system. The above problem reduces (see appendix I) to a modular subset sum (type problem); a problem known to be NP complete. As explained in appendix H

$$\begin{aligned} |\epsilon_{p2}| &= O(2^{-z}). \\ &\text{or} \\ |\epsilon_{p2}| &= O((2^{-z})(\log n/2^n)) \end{aligned}$$

will be two possibilities for function for ϵ_{p2} for the separation to work as explained in appendix H. ϵ_{p2} mean episilon for problem 2.

How to Use the Oracles as Part of a Quantum Classical Hybrid Algorithm

Firstly a sufficiently large $u \in \mathbb{Z}^+$ is chosen for linear scaling constraints on the coefficients n_j , such that $n_j \in (2j - 1) * u, 2j * u]$. Then the n_j are searched for with the possible bounds in some way that test all the possibilities in these exponential bounds. In informal description the oracles "are internal" and an "invisible oracle" (no external circuit required see proof in appendix A) which is the "opposite" to the usual "external subcircuit oracle" for some circuit. A quantum oracle algorithm is repeatedly used, over the search space, to evaluate each possible solution, in polynomial time on a quantum computer, for if a solution is a solution to one of two types type of modular system (one modular system is equal to zeros or near zeroes for the equations the other is equal to some rational multiples of pi for the equations); the above two quantum oracles that

are used there are our quantum algorithm 1 and quantum algorithm 2. Because quantum algorithms are probabilistic we would need to repeat the quantum algorithm a number of times for a single potential solution and do this number of repeats for all potential solutions in the search space for solutions. If the above process does not find a solution then the process is repeated with a larger value of u to increase the sizes of the exponential bounds for each.

The Number Theoretic Problem Solved by Algorithm 1

The power of quantum computation is frequently defined by its ability to extract global properties from exponential Hilbert spaces, most notably in period-finding and unstructured search. However, a large class of problems involving irrational rotations and Diophantine approximations remains under-explored in the context of quantum advantage. We address the problem of verifying a solution vector $n = (n_1, \dots, n_n)$ for a system of $M = 2^n$ modular equations:

$$\left| \left(\sum_{j=1}^n n_j \gamma'_j x_j^{(i)} \right) - \alpha_i \right| \pmod{1} < \epsilon, \quad \forall i \in \{0, \dots, 2^n - 1\}$$

where γ'_j are known irrational constants and α_i are structured offsets. Classically, verifying such a system is strictly $O(2^n)$ as it requires the independent evaluation of each subset sum.

The Kronecker Structural Isomorphism

We address the problem of verifying a solution vector $n = (n_1, \dots, n_n)$ for a system of $M = 2^n$ modular equations. Crucially, the structure of this system is not arbitrary; it is inherited directly from the Kronecker tensor product structure of the n -qubit Hilbert space. The system is defined such that the coefficients $x_j^{(i)}$ correspond to the bit-string representations of the indices $i \in \{0, \dots, 2^n - 1\}$. Specifically, the i -th equation is a subset sum determined by the binary expansion of i :

$$\left| \left(\sum_{j=1}^n n_j \gamma'_j \mathbb{I}(i, j) \right) - \alpha_i \right| \pmod{1} < \epsilon$$

where $\mathbb{I}(i, j)$ is the j -th bit of the index i . This mapping ensures that the system of equations is isomorphic to the phase accumulation of a quantum state under local, independent rotations.

One purpose of this paper is to show there is an interesting intrinsically quantum way (though its part of a quantum-classical hybrid algorithm too) to solve the Weyl systems of equations that is conceptually different from the purely classical brute force method presented in this paper too. Another purpose of

this paper is we claim our quantum algorithm, that involves an $O(2^n)$ number of solutions is faster/has better time complexity (polynomial time) than compared to any known classical algorithm in our ACS framework. Our paper introduces a new mapping that shows how a Weyl system of equation can be implemented on a quantum computer using polynomial resources.

The purpose of this paper is to present "a quantum mapping/a type of primary build block which we may informally refer to as a "primitive" " .

Where we compared to classical everywhere in our paper means we are referring to the specific brute force type classical algorithm presented below as a baseline in our paper, and it also means we claim our quantum algorithm is faster than any other possible classical algorithm in other words we do not think the general Weyl systems of equations can be solved or verified classically in polynomial time and polynomial space complexity (this is expressed by our conjecture 1). Comparing to this specific brute force algorithm illustrates the conceptually quantum intrinsic algorithms for solving exactly the same problem.

2. The Quantum Intrinsic Parallelism

the quantum processor is intensional. By applying a local rotation $R_j = e^{2\pi i n_j \gamma'_j}$ to the j -th qubit, the laws of tensor products automatically generate the entire 2^n set of subset sums in the state's phases:

$$|\psi\rangle = \bigotimes_{j=1}^n \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i n_j \gamma'_j} |1\rangle) = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} e^{2\pi i (\sum n_j \gamma'_j x_j^{(i)})} |i\rangle$$

The quantum computer does not "calculate" the 2^n sums; it embodies them. The speedup arises because the quantum state naturally maintains the Kronecker structure as a single coherent object, whereas a classical computer must break that structure apart into 2^n independent strings of data to process them.

The Classical Baseline

The classical algorithm is tasked with verifying a candidate solution n against a system where the coefficients $\chi_j^{(i)}$ are explicitly derived from the Kronecker tensor product structure. We use the classical brute force type algorithm as a baseline to show that the classical verification part can be sped up with a quantum computer for this specific classical algorithm which illustrates/evidences our quantum algorithm is conceptually different from this brute force type classical algorithm.

1. The Kronecker-Isomorphic System

The system of $M = 2^n$ equations is structured as follows:

$$\left| \left(\sum_{j=1}^n n_j \gamma'_j \chi_j^{(i)} \right) - \alpha_i \right| \pmod{1} < \epsilon$$

Here, the matrix of coefficients X (where $X_{ij} = \chi_j^{(i)}$) is the binary matrix representation of the n -fold Kronecker product of the basis vectors. This structure is precisely what allows the quantum computer to prepare the state using n independent local rotations.

Conjecture 1: Intractability of Classical Weyl Verification

We operate under the structural conjecture that no classical deterministic or bounded-error probabilistic algorithm can verify the global null-congruence of the 2^n Weyl equations in polynomial time.

Justification: Because the base coefficients γ'_j are continuous irrational constants, their fractional transitions modulo 1 are governed by Weyl's Equidistribution Theorem. Consequently, the modular phase errors δ_x behave pseudo-randomly across the 2^n vertices of the Boolean hypercube. Since the non-linear modulo 1 reduction destroys algebraic distributivity, the system cannot be factored or localized. To guarantee the maximum error condition $\max |\delta_x| < \epsilon$ holds globally without a single violating path, a classical machine is mathematically forced to extensionally compute the path space, imposing an $\mathcal{O}(2^n)$ verification bottleneck.

If this conjecture 1 in the box above holds, it provides an explicit structural separation between the complexity classes MA and QCMA, in the ACS sense, demonstrating that there exists a language verifiable by a quantum proof that is not efficiently verifiable by a classical randomized proof.

While the problem above we define does not formally require P not equal to NP, its classical hardness mirrors the intuition of P not equal to NP. Specifically, if a polynomial-time classical algorithm existed for the Weyl-system verification, it would give evidence for the existence of an algebraic shortcut that collapses the pseudo-random distribution of the "Weyl manifold" a result that would contradict currently held views on the hardness of high-dimensional modular systems.

Complexity separations require a concrete, mathematically rigorous problem structure to analyze. Just as Shor's algorithm established a quantum-classical separation solving specialisations of the Hidden Subgroup Problem, any complexity analysis must investigate a well-defined architectural or algebraic structure. The ACS framework and the high-dimensional Weyl modular system are precise mathematical structures that isolate the exact bottleneck, non-separable Kronecker phase accumulation, that separates MA from QCMA. The role of the

Weyl problem is similar to the "solving specialisations of the Hidden Subgroup Problem" above.

When we refer to classical Algorithm comparison throughout this paper we are comparing to the brute force type classical algorithm described below as a concrete example; we do not think there is a polynomial time classical algorithm for the problem (a expressed by conjecture 1).

3. The Classical Verification Algorithm

Objective: Verify n against $M = 2^n$ equations. Despite the Kronecker rule, classical verification is strictly extensional.

- **Algorithm Steps:**

1. **Iterate:** $i = 0 \rightarrow 2^n - 1$.
2. **Row Generation:** Construct X_i (the i -th row of the n -fold Kronecker product binary matrix).
3. **Point-wise Evaluation:** Compute subset sum $S_i = \sum_{j=1}^n n_j \gamma_j \mathbb{I}(i, j)$.
4. **Modular Check:** Evaluate $\delta_i = |S_i - \alpha_i| \pmod{1}$.
5. **Termination:** Fail if any $\delta_i \geq \epsilon$.

$\mathbb{I}(i, j)$ is an indicator function that is 1 or 0 depending on if the i th bit of j is 1 or 0. The Weyl Criterion ensures that if the solution is wrong, the errors δ_i will be distributed across the circle, requiring the classical algorithm to potentially check a significant fraction of the 2^n equations before finding a violation.

The Quantum Speedup: Exploiting the Tensor Isomorphism

Importantly the **Quantum Algorithm** uses the Kronecker structure of Hilbert space as a direct physical mirror of the mathematical problem.

- **Classical Perspective:** No known classical algorithm uses the Kronecker structure to do the verification in polynomial time and polynomial space.
- **Quantum Perspective:** The Kronecker structure is the natural geometry of the qubits. By applying n local gates, the quantum computer "calculates" all 2^n subset sums simultaneously through state evolution.

Summary of Complexity Gap: (Compared to above Classical Algorithm)

The Kronecker structure provides the **Quantum Algorithm** with an $O(n)$ path to state preparation (the LHS), which is then resolved by a $O(n^2)$ QFT. The above **Classical Algorithm**, despite knowing the Kronecker rule, is in $O(2^n)$ time because it cannot perform the global interference required to check the "offsets" (α_i) without visiting every single coordinate in the exponential space and it needs to check an exponential number $O(2^n)$ constraint inequalities " $\max_j |\theta_j - \frac{1}{2}| < \epsilon$ " " $\max |\delta_x| < \epsilon$ " described in the "problem solved" sections above for algorithms 2 and algorithms 1 respectively.

This speedup is due to the **quantum data structure (the Wavefunction)** in handling tensor-product- based modular system.

This **demonstrates that the verification of high-dimensional Weyl modular systems provides an Architectural Complexity Separation (ACS) between MA and QCMA.**

The Role of Repetition

Because these are probabilistic quantum algorithms, they must be **repeated K times** to ensure the recovered period P is not a false positive. In Algorithm 1, if the known period of the anchor function consistently appears after K shots, the solution is verified. If the period "breaks" into Weyl-distributed noise, the solution is rejected.

Note that K is of polynomial size and takes polynomial space to recover P (recovered in polynomial time, and this is implied from Shor's algorithm as mentioned below).

This framework proves that by utilizing the Kronecker tensor structure, a quantum computer can verify an exponential system (2^n equations) in polynomial time ($O(n^2)$) by sensing the restoration of global phase stability.

Quantum Algorithm 1

To achieve maximum density, we define the **Kronecker-isomorphic quantum verification** framework for $M = 2^n$ equations.

Objective: Verify $(\sum n_j \gamma_j \chi_j^{(i)}) - \alpha_i \equiv 0 \pmod{1}$.

- **Operator Construction:** The state $|\psi\rangle$ is prepared as a periodic anchor with known period P . We apply $U = B \cdot A$:

$$(A)_{ii} = \exp \left(2\pi i \sum_{j=1}^n n_j \gamma_j \chi_j^{(i)} \right), \quad (B)_{ii} = \exp(-2\pi i \alpha_i)$$

- **The Generalized QFT Equation:** The standard QFT factor $\exp(-2\pi i \frac{xy}{2^n})$

is multiplied “beside” the modular error δ_i :

$$P(|0\rangle) = \left| \frac{1}{2^n} \sum_{i=0}^{2^n-1} e^{2\pi i \delta_i} \right|^2 \quad \text{where } \delta_i = (LHS_i - \alpha_i)$$

- **Convergence:** If n is a solution, $\delta_i \rightarrow 0$, $e^{2\pi i \delta_i} \rightarrow 1$. The sum converges to unity, restoring the **known period** P .
- **Divergence:** If incorrect, δ_i follows the Weyl Criterion, causing destructive interference; $P(|0\rangle) \approx 1/2^n$.

1. The Periodic Anchor (Preparation)

Before applying the modular system, the quantum register is prepared in a state $|\psi_p\rangle$ with a **known period** P (e.g., via modular exponentiation $x^a \pmod{N}$). In a clean QFT, this state would yield sharp spectral peaks at frequencies k/P .

2. The Phase "Mask" (Algorithm 1 Execution)

The modular system $(LHS_i - \alpha_i)$ is applied as a phase-mask **beside** the existing periodic amplitudes. The state becomes:

$$|\psi_{mask}\rangle = \frac{1}{\sqrt{M}} \sum_r e^{2\pi i(\delta_r P + b_0)} |rP + b_0\rangle$$

where $\delta_i = LHS_i - \alpha_i$ is the modular error of the i -th equation.

3. Convergence to Known QFT

- The relationship to periodic function is determined by the interference sum
- **If n is a Solution:** $\delta_i \approx 0 \pmod{1}$ for all i . The phase multiplier $e^{2\pi i \delta_i} \approx 1$. This restores the original periodic symmetry.
- **If n is a Non-Solution:** The irrational δ_i acts as a “wobble” that violates the periodic condition. The phase factors $e^{2\pi i \delta_i}$ are randomly distributed (Weyl Criterion). This causes destructive interference, “breaking” the QFT and preventing the recovery of the period P .

The period P is recovered using a Shor type continued fraction algorithm, and hence the recovery of the period P is in polynomial time and polynomial space due to Shor.

4. Verification via Repetition

Because the measurement is probabilistic, the algorithm is repeated K times:

- **Success:** If the known period P is recovered consistently, the solution is verified.
- **Failure:** If the output is random noise (Weyl distribution), the solution is rejected.

In summary, Algorithm 1 doesn’t just check if $\delta_i = 0$; it checks if the known periodic signal is preserved. The QFT is the “filter” that only lets the signal through if the solution n has successfully canceled the irrational noise.

To understand how the “phase mask” is embedded, we must look at the Kernel of the summation. In a standard QFT, the probability amplitude for a state $|y\rangle$ is determined by the kernel $\exp(-2\pi i \frac{xy}{2^n})$. In Algorithm 1, our modular system is inserted as a **complex multiplicative factor** that sits directly “beside” that standard term.

The Embedded QFT Equation for Algorithm 1

If we apply Algorithm 1 to a periodic anchor state $|\psi_p\rangle$, the probability amplitude $A(y)$ for measuring frequency y is:

$$A(y) = \frac{1}{2^n} \sum_{x=0}^{2^n-1} \left[\underbrace{e^{2\pi i(\sum n_j \gamma_j \chi_j^{(x)} - \alpha_x)}}_{\text{The Phase Mask (LHS - RHS)}} \right] \cdot \underbrace{e^{-2\pi i \frac{xy}{2^n}}}_{\text{Standard QFT Factor}}$$

with δ_x notation:

$$A(y) = \frac{1}{2^n} \sum_{x=0}^{2^n-1} e^{2\pi i \delta_x} \cdot e^{-2\pi i \frac{xy}{2^n}}$$

How this "Mask" Controls Convergence

The “mask” acts as a **gatekeeper** for the constructive interference:

1. 1. When n is a Solution:

The phase mask $e^{2\pi i \delta_x}$ becomes $\approx e^0 = 1$ for all x . The equation collapses back to the **Standard QFT equation**. Because the input was a periodic function with known period P , the sum undergoes constructive interference at $y \approx k \frac{2^n}{P}$. You “see” the period because the mask became transparent.

1. **When n is NOT a Solution:** The mask $e^{2\pi i \delta_x}$ is a sequence of irrational, pseudo-random phases (Weyl noise). Even if the underlying signal is periodic, these extra factors rotate each vector in the sum to a random angle. The absolute value $|A(y)|$ collapses because the vectors no longer point in the same direction. The "standard QFT Factor" cannot find a frequency because the mask has destroyed the signal's coherence.

Matrix Entry Relation

In the matrix $U = B \cdot A$, the entry U_{xx} is exactly that first bracketed term:

$$U_{xx} = \exp(2\pi i \delta_x)$$

The QFT then takes these diagonal entries and “diffuses” them across the basis. If those diagonal entries are all 1 (the identity matrix), the QFT behaves normally. If they are “wobbly,” the QFT output is noise.

To be precise, in problem 1, the δ_i in problem one has to satisfy being inside one of two finite intervals separated by a "finite promise gap" (this is shown in section below along with using the theorem in the Appendix G) where one interval corresponds to the anchor period being recovered successfully if δ_i is in the above interval; and the other interval corresponds to the anchor period recovered unsuccessfully if δ_i is in the above interval. A similar above comment applies to problem 2 "for the epsilons".

Statement of Problem Solved by Algorithm 1

For Algorithm 1, we define a system of $M = 2^n$ modular equations where the objective is to verify if a candidate vector $n \in \{0, 1\}^n$ satisfies the null-congruence $\delta_i \approx 0 \pmod{1}$.

1. The Core Operators

The verification is performed by the unitary operator $U = B \cdot A$, where:

- **Operator A (LHS Matrix):**

$$A = \bigotimes_{j=1}^n R_j, \quad R_j = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i n_j \gamma'_j} \end{pmatrix}$$

Via the Kronecker product, the diagonal entries are:

$$(A)_{ii} = \exp \left(2\pi i \sum_{j=1}^n n_j \gamma'_j \mathbb{I}(i, j) \right)$$

This "embodies" the Left-Hand Side subset sums for all 2^n equations simultaneously.

- **Operator B (RHS/Target Matrix):**

$$B = \text{diag}(e^{-2\pi i \alpha_0}, e^{-2\pi i \alpha_1}, \dots, e^{-2\pi i \alpha_{M-1}})$$

This encodes the target offsets α_i (the Right-Hand Side).

2. The QFT Generalization Equation

The algorithm applies U to a periodic anchor state $|\psi_p\rangle$. The probability amplitude $A(y)$ for measuring frequency y is a generalization of the standard QFT, where the modular error δ_i is embedded "beside" the Fourier factor.

2.1 The "Promise Gap" for Algorithm 1

According to complexity theory, a formal solution and verification protocol for a quantum verification class like QMA or QCMA (and the separation between them) must satisfy the standard promise problem verification framework. A valid solution must meet the following criteria :

- **Promise Problem Partition:** The input space is divided into two disjoint sets, YES instances (A_{yes}) and NO instances (A_{no}), where no valid input can belong to both.
- **Polynomial-Size Proof (Witness):** The all-powerful prover (Merlin) provides a proof whose size is strictly bounded by a polynomial of the input length n (a classical string for QCMA, or a quantum state for QMA).
- **Efficient Quantum Verifier (Arthur):** Arthur executes a polynomial-time uniform quantum circuit (or a polynomial-time quantum algorithm).
- **Completeness ($a \geq c$):** For any YES instance, there exists at least one proof that convinces Arthur to accept with a high probability (e.g., $\geq \varsigma$ for some positive constant $\varsigma > 0$).
- **Soundness ($b \leq s$):** For any NO instance, every possible proof submitted by a cheating prover causes Arthur to reject with high probability, meaning the maximum acceptance probability is bounded below a lower threshold (e.g., $\leq \varpi$ for some positive constant $\varpi > 0$).
- **Measurable Promise Gap:** There is a distinct, non-vanishing separation between the completeness and soundness thresholds ($c - s \geq \frac{1}{\text{poly}(n)}$), which can be robustly amplified using a polynomial number of repetitions ($K = \text{poly}(n)$) paired with standard concentration bounds.

The above applied to the Architectural Complexity Framework

In appendix G we analyse a noisy version of the QFT which applies as follows

1. Promise Problem Partition

- **In WSOT context:** our paper defines the architectural decision problem by dividing the input space into two disjoint categories based

on physical behavior: YES instances (where the periodic signal and anchor period remain stable, $S_n = O(1)$) and NO instances (where noise dominates and phase coherence collapses, $S_n \rightarrow \infty$). Note that the matrices required to formulate the problem (A, B etc) which are required by quantum Arthur to verify need not be constructed efficiently (or in polynomial time by quantum Arthur or quantum Merlin) as it is only the task for Arthur is to verify in polynomial time and polynomial space with a quantum algorithm.

- **The Theorem’s Role:** our theorem in Appendix F mathematically formalizes this exact boundary, proving a sharp threshold that cleanly partitions the instance space without requiring a sluggish intermediate transition zone.

2. Polynomial-Size Proof (Witness)

- **In WSOT:** In the context of our complexity separation (QCMA framework), Merlin provides a compact, polynomial-sized proof (such as the classical description of the anchor period or phase configuration).
- **The Theorem’s Role:** The theorem guarantees that this polynomial-size information is sufficient because the verification outcome depends entirely on the robust scaling of the noise budget S_n , which scales cleanly with the input size n rather than requiring exponentially large data.

3. Efficient Quantum Verifier (Arthur)

- **In WSOT context:** Arthur executes a polynomial-time uniform quantum circuit to measure and verify the phase coherence and signal recovery.
- **The Theorem’s Role:** The mathematical structure proved in our theorem maps directly onto the quantum operations Arthur performs, ensuring the verification circuit runs in polynomial time ($\text{poly}(n)$).

4. Completeness ($a \geq c$)

- **In WSOT context::** For any valid (YES) instance where the noise budget is bounded ($S_n = O(1)$), Arthur must accept with high probability.
- **The Theorem’s Role:** our theorem explicitly guarantees this via the lower bound ($P(y \in G_r) \geq \frac{4}{\pi^2} e^{-0.4S_n} > 0$), proving that a valid configuration maintains a strictly positive, non-vanishing success probability.

5. In WSOT context: ($b \leq s$)

- **The Integration:** For any invalid (NO) instance where noise overwhelms the system ($S_n \rightarrow \infty$), any cheating prover must be rejected with high probability.
- **The Theorem's Role:** The theorem proves that the success probability vanishes asymptotically ($P(y \in G_r) \rightarrow 0$), ensuring that invalid instances cannot trick Arthur into a false acceptance.

6. Measurable Promise Gap & Amplification

- **In WSOT context:** Complexity theory requires a clear, non-vanishing gap between the completeness and soundness thresholds that can be boosted reliably.
- **The Theorem's Role:** Because our theorem establishes a sharp threshold between the constant-bounded YES case and the vanishing NO case, it creates a pristine probabilistic separation. This allows Arthur to use a polynomial number of repetitions ($K = \text{poly}(n)$) paired with concentration bounds to amplify the confidence to an arbitrarily high certainty, satisfying the strict requirements of the complexity class separation.

To formally establish the architectural complexity separation, we set $A_z = \delta_x$ and invoke the sharp threshold theorem proven in this appendix G. We use the noisy version of the QFT, in appendix G, the noise term in each individual qubit is contributed by (using ϵ_{p1} or ϵ_{p2}), or its role is taken by, a form of a Weyl equation as expressed in algorithm one and algorithms two. This enables Arthur today a polynomial time verification; the noise in the QFT is linked to the residue modulo 1 (or $2\pi i$). By mapping the physical noise budget scaling S_n directly to the verifier's completeness and soundness criteria, the theorem guarantees a rigorous dichotomy: for valid (YES) instances where $S_n = O(1)$, the verification success probability is bounded below by $P(y \in G_r) \geq \frac{4}{\pi^2} e^{-0.4S_n} > 0$, whereas for invalid (NO) instances where $S_n \rightarrow \infty$, the acceptance probability vanishes as $P(y \in G_r) \rightarrow 0$. This sharp transition satisfies the promise problem definition required to separate QMA, in the ACS sense, as it creates a pristine probabilistic separation with an explicit quantitative gap size defined by the completeness lower bound of $\frac{4}{\pi^2} e^{-0.4S_n}$ versus the soundness limit approaching 0. Consequently, employing a polynomial number of repetitions $K = \text{poly}(n)$ allows Arthur to amplify this exact separation and reliably distinguish valid instances from invalid ones with exponentially small error. The proof in appendix I is quantum Arthur's verification algorithm which is used to separate of QMCA and QMA in the ACS framework. We already showed there is no polynomial time algorithm for the Weyl problem in problem 2 as shown in appendix G by reducing it to the subset sum problem.

Description of Algorithm 2

In the Latent QFT framework of Algorithm 2, the α constants are not just random offsets; they are the Phase Aligners that allow the Kronecker product

to "snap" into the QFT structure.

1. The Alpha Matrix (B)

Because Algorithm 2 relies on the n -fold tensor product to generate the 2^n equations, the α values must also follow a specific structure to avoid breaking the tensor factoring.

We define the Target Phase Matrix B as a diagonal matrix:

$$B = \text{diag}(e^{-i\alpha_0}, e^{-i\alpha_1}, \dots, e^{-i\alpha_{M-1}})$$

To maintain the "Latent QFT" property, this matrix B is itself a Kronecker product of n local alpha-shifts:

$$B = \bigotimes_{j=1}^n \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\alpha_j} \end{pmatrix}$$

This ensures that the target side of our 2^n equations "involving the α_i " are actually combinations of n base α_j constants.

3. The Mechanism of the Latent QFT

The interaction follows this logical chain:

1. **Phase Subtraction:** The matrix B effectively "shifts" the irrational γ'_j values.
2. **Target Alignment:** For the j -th qubit, we are looking for an n_j such that:

$$n_j \gamma'_j - \alpha_j \rightarrow \frac{2\pi}{2^j}$$

3. **The Emergence:** If every local factor R_j hits its rational target, the global matrix $U(n)$ becomes the QFT matrix.

4. Why this matters for the 2^n equations

When we expand $U(n)$, the i, j entry of the global matrix contains the combined term:

$$\exp\left(i \left[\left(\sum \text{subset of } n_j \gamma'_j \right) - \left(\sum \text{subset of } \alpha_j \right) \right] \right)$$

If the solution is correct, every one of these 2^n entries simultaneously aligns with the exact complex phase required by the Standard QFT.

Summary: The α constants act as the coordinate shift that moves our irrational system onto the QFT's rational grid. The interaction is a Local-to-Global mapping: we adjust n local phases $(n_j \gamma'_j - \alpha_j)$, which, via the Kronecker product, automatically corrects the 2^n global phases of the latent manifold.

To express $n_j\gamma'_j - \alpha_j \rightarrow \frac{2\pi}{2^j}$ precisely, we use the binary indicator function $\mathbb{I}(i, j)$ to show how the i -th diagonal entry of the global operator $U(n)$ is constructed from the n local tensor factors.

1. The Precise Entry Definition

The i -th diagonal entry (where $i \in \{0, \dots, 2^n - 1\}$) of the expanded Kronecker product $U(n) = \bigotimes_{j=1}^n R_j$ is given by:

$$(U)_{ii} = \exp \left(i \sum_{j=1}^n \mathbb{I}(i, j) \cdot (n_j\gamma'_j - \alpha_j) \right)$$

2. The Latent Mapping to the QFT

For the operator $U(n)$ to converge to the Standard QFT, each local phase $(n_j\gamma'_j - \alpha_j)$ must satisfy the specific rational requirement of the j -th qubit's contribution to the Fourier transform.

In a standard QFT, the i -th phase is $\frac{2\pi \cdot i \cdot k}{2^n}$. For latent version, the convergence happens when:

$$n_j\gamma'_j - \alpha_j = \frac{2\pi}{2^j}$$

When this condition is met for all n factors, the sum in the exponent becomes:

$$\sum_{j=1}^n \mathbb{I}(i, j) \frac{2\pi}{2^j} = 2\pi \sum_{j=1}^n \frac{\text{bit}_j(i)}{2^j}$$

3. Why this equals the "Correct Rational Multiple of π "

The term $\sum \frac{\mathbb{I}(i,j)}{2^j}$ is exactly the binary fraction representation of the index i .

- Let i in binary be $b_1b_2 \dots b_n$.
- The sum is $0.b_1b_2 \dots b_n$ in base 2.
- Therefore, $(U)_{ii} = e^{2\pi i(\text{frac}(i))}$.

This is precisely the value required in the i -th position of a QFT matrix to perform a frequency rotation. The " π " emerges because the solution to our system forces the irrational subset sum to align with the natural geometry of the 2^n -th roots of unity.

4. The "Structural Snap"

If n is off by even a small amount, the Weyl Criterion takes over. Because γ'_j is irrational, the error doesn't stay small; it distributes the phases $e^{i\delta}$ across the circle.

The matrix only "snaps" from a collection of random complex numbers into the highly ordered QFT manifold when the candidate n_j integers perfectly bridge the gap between the irrational γ'_j and the rational grid $2\pi/2^j$. At that moment, the 2^n equations are solved simultaneously because they are all shadows of the same n tensor factors.

Algorithm 2: Latent QFT via Structural Isomorphism

Algorithm 2 solves the **Rational-Manifold System**: identifying a vector of dimension n that transforms an irrational tensor space into a rational Fourier manifold.

1. Core Operator Construction

The operator $U(n)$ is an n -fold Kronecker product of "wobbly" Hadamard-like factors. The interaction with the problem's α -constants occurs at the local qubit level:

$$U(n) = \bigotimes_{j=1}^n R_j, \quad R_j = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & \exp(i\theta_j) \end{pmatrix}$$

Where the **Local Phase Component** is:

$$\theta_j = (n_j \gamma'_j - \alpha_j) \pmod{2\pi}$$

2. Matrix Entry Mapping & The QFT Limit

The global matrix entries $(U)_{ii}$ are subset sums of these local components. The **Latent QFT** emerges as the limit of these phases approaching the harmonic frequencies of the Hilbert space:

$$(U)_{ii} = \exp \left(i \sum_{j=1}^n \mathbb{I}(i, j) \cdot \theta_j \right)$$

As $n \rightarrow n^*$, the local phases satisfy the **Rational Mapping Condition**: $\theta_j \rightarrow \frac{2\pi}{2^j}$. The global entry then converges to the standard Fourier phase ϕ_i :

3. Convergence: The "Non-Zero" Solve

- **Algorithm 1 (Zero-Solve):** Seeks $\theta_j \approx 0 \pmod{2\pi}$, making $U \rightarrow I$ (Identity).
- **Algorithm 2 (Non-Zero Solve):** Seeks $\theta_j \approx$ Rational Multiple of π . Success is not the absence of phase, but the presence of a **perfect rational gradient**. The operator $U(n)$ is a "refractive lens" that only focuses the signal when the integer set n aligns the irrationality.

4. Search Dynamics & Detection

The algorithm navigates the search space by leveraging the **interference pattern** of a periodic anchor state $|\psi_P\rangle$ with known period P .

1. **State Preparation:** Apply $U(n)$ to $|\psi_P\rangle$.

$$U_{i,j} = \frac{1}{\sqrt{2^n}} e^{i \cdot \delta_{ij}} = \frac{1}{\sqrt{2^n}} e^{i \cdot \sum_{k=0}^{n-1} \mathbb{I}(i,j,k) \cdot (n_k \gamma_k - \alpha_k)}$$

2. **Probability Amplitude:** The frequency-space amplitude $A(y)$ follows the QFT equation:

Let the input anchor state be expressed as a linear combination of basis states:

$$|\psi_P\rangle = \sum_{j=0}^{2^n-1} c_j |j\rangle$$

Where c_j represents the initial amplitude of each state $|j\rangle$ in the periodic anchor function. When the operator U acts on this state, it performs a matrix-vector multiplication. Using our exact matrix entry definition for $U_{i,j}$, the probability amplitude for measuring the specific ket $|v\rangle$ is given by:

$$\text{ProbAmp}(v) = \langle v | U | \psi_P \rangle = \sum_{j=0}^{2^n-1} U_{v,j} \cdot c_j$$

substituting our precise Weyl Kronecker phase definition $U_{v,j}$ yields the final amplitude

$$\text{ProbAmp}(v) = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} c_j \cdot e^{i \cdot \sum_{k=0}^{n-1} \mathbb{I}(v,j,k) \cdot (n_k \gamma_k - \alpha_k)}$$

the Latent QFT Target Symmetry This equation shows exactly how the system "snaps" into focus

- **Incorrect n_k :** The phases inside the exponent are Weyl-distributed. They scramble the incoming amplitudes c_j destructively across the manifold, causing $\text{ProbAmp}(v)$ to collapse into a flat noise floor for all values of v .
- **Correct n_k (The Refractive Lens):** The local integers perfectly align the irrationalities. The exponent simplifies such that the matrix elements $U_{v,j}$ smoothly match the exact geometric steps of a clean Fourier transform closely enough.

Because U has now crystallized into the QFT matrix, evaluating $\text{ProbAmp}(v)$ executes a valid Fourier transform on the periodic coefficients c_j . This causes massive constructive interference, yielding sharp, high-amplitude peaks precisely at the values of v that correspond to the hidden period P of our anchor state.

The QFT is truly latent, the second term -the "Standard QFT Basis" is not explicitly hard-coded with a π in the algorithm's execution. Instead, the entire interference pattern emerge from the interaction between the operator and the state.

Here is the "pi-less" formulation for the latent emergence:

1. The Latent Amplitude Equation

The probability amplitude $A(y)$ is generated by applying the "wobbly" operator $U(n)$ to the anchor state $|\psi_P\rangle$. The QFT is not "added"; it is the **resulting state of the summation**:

$$A(y) = \frac{1}{2^n} \sum_{i=0}^{2^n-1} \exp \left(i \left[\sum_{j=1}^n \mathbb{I}(i, j) \theta_j \right] \right) \cdot \langle y | i \rangle$$

In this form, the "Standard QFT" does not exist yet. There is only a candidate phase rotation and the natural basis of the Hilbert space.

2. The Geometric Convergence (The Emergence of π)

The π constant emerges as a **geometric limit** when the search hits the solution n^* . We define the convergence as follows:

$$\lim_{\theta_j \rightarrow \frac{2\pi}{2^j}} A(y) = \underbrace{\frac{1}{2^n} \sum_{i=0}^{2^n-1} e^{2\pi i (\text{frac} \theta_i)} \cdot e^{-2\pi i \frac{iy}{2^n}}}_{\text{Classical QFT Definition}}$$

- **Before Convergence:** The exponent is just an irrational sum of θ_j . The 2π is "latent" "it is the circumference of the unit circle that the irrational phases are traversing.
- **At Convergence:** The system hits the **Rational Resonance**. The term $\sum \mathbb{I}(i, j) \theta_j$ becomes a perfect fraction of the circle. Only at this point does the expression mathematically collapse into the standard QFT equation we recognize.

The convergence has to match the standard QFT **close enough** it does not need to be perfectly equal to the QFT for algorithm 1 or algorithm 2 to work.

The "Promise Gap" for Algorithm 2

Problem 2 like problem 1, the δ_i in problem one has to satisfy being inside one of two finite intervals separated by a "finite promise gap" (this is shown in Appendix H-which is not-similar the corresponding proof for algorithm 1) where one interval corresponds to the anchor period being recovered successfully if δ_i is in the above interval; and the other interval corresponds to the anchor period recovered unsuccessfully if δ_i is in the above interval. A similar above comment applies to problem 2 "for the epsilons". The proof in appendix H is quantum

Arthur's verification algorithm which is used to separate of QMCA and QMA in the ACS framework. We already showed there is no polynomial time algorithm for the Weyl problem in problem 2 as shown in appendix I by reducing it to the sub set sum problem.

3. Search Space: Navigating the Weyl Noise

The algorithm finds the integers n by monitoring the **Symmetry Collapse**:

1. **Weyl Phase (Incorrect n)**: The phases θ_j are "off-grid." By the Weyl Criterion, the sum $\sum e^{i\delta_i}$ behaves like a random walk on the complex plane. The amplitude $A(y)$ remains near zero for all y .
2. **Crystallization (Correct n)**: As θ_j approaches the rational multiples, the random walk straightens into a coherent circle (a periodic wave).
3. **The Detector**: Because $|\psi_P\rangle$ has a known period, it acts as a "tuning fork." If $U(n)$ has successfully become a QFT, it will perfectly "ring" with the period P .

4. Summary of the "Non-Zero" Solve

frac_i represents the **rationalized limit** of $\delta_i(n)$

- **Algorithm 1**: Finds $\sum \theta_j \rightarrow 0$
- **Algorithm 2**: Finds $\sum \theta_j \rightarrow \text{Harmonic Ratios}$

The " π " is never written into the code of Algorithm 2; it is the **frequency** at which the quantum computer finally "hears" the anchor function's period. If you don't find the right n_j , the "pi" never appears, and the period P remains hidden in Weyl noise.

Another Expression of Algorithm 2

2. The Latent QFT Convergence (The Pi-Less Limit)

The "Standard QFT" is defined by the goal of the search. Convergence is reached when the local phases θ_j align with the natural harmonic divisions of the Hilbert space.

The Solution Condition:

$$n \text{ is a solution iff } \theta_j = \text{Harmonic}_j$$

As the search reaches the solution n^* , the operator $U(n)$ undergoes a structural isomorphism:

$$\lim_{\theta_j \rightarrow \text{Harmonic}_j} U(n) \equiv \text{QFT}_{2^n}$$

In this limit, the i -th entry of the matrix $(U)_{ii}$ converges to the exact phase required to rotate the i -th basis state into the frequency domain. The constant π is not hard-coded; it is the **geometric byproduct** of the phases reaching a perfect rational gradient across the 2^n equations.

3. Search Dynamics and Detection

The algorithm navigates the n -dimensional search space by monitoring the **Global Resonance** of a periodic anchor function $|\psi_P\rangle$.

- **Step 1: Superposition Testing:** The circuit applies the candidate $U(n)$ to the anchor $|\psi_P\rangle$.
- **Step 2: The Amplitude Equation:**

$$A(y) = \frac{1}{2^n} \sum_{i=0}^{2^n-1} (U)_{ii} \cdot \langle y|i\rangle$$

- **Step 3: Symmetry Restoration vs. Weyl Noise:**
 - **Incorrect n :** The phases $\sum \mathbb{I}(i,j)\theta_j$ are **Weyl-distributed**. They behave as a random walk on the unit circle, leading to total destructive interference. The anchor's

period P is destroyed, leaving only a $1/2^n$ noise floor.

- **Correct n :** The system hits the **Rational Resonance**. $U(n)$ "crystallizes" into a QFT and performs a valid frequency transform. The periodic signal "shines through," resulting in sharp probability peaks at the known period P .

4. The "Non-Zero" Solve Distinction

- **Algorithm 1:** Solves for the **Null State** ($\theta_j \rightarrow 0$), where the operator becomes the Identity (I).
- **Algorithm 2:** Solves for the **Ordered State** ($\theta_j \rightarrow$ Rational Grid), where the operator becomes the Fourier Transform.

Result: Algorithm 2 finds the integers n by seeking the point where the "chaos" of irrational subset sums collapses into the periodic rational manifold. Success is detected not by the absence of signal, but by the **restoration of a known global symmetry**.

Complexity Analysis

To analyze the complexity, we define n as the number of variables (qubits) and W as the search range for each integer n_j (i.e., $n_j \in [0, W]$). The total search space for the vector n is *Note that we would increase u if j increased, we refer to u as the winding number, "the bigger u becomes then the bigger some n_j becomes, & hence the longer a solution vector becomes-in other words the solution is not the shortest vector here". Note the two different Weyls systems of algorithm 1 and 2, which we refer to as a Weyl System of of Type 1 (WSOT1) and a Weyl System of of Type 2 (WSOT2) respectively, both naturally have more than one solution each*

Input Parameters and Search Ranges

Parameter	Definition	Search Range
n	Number of variables (qubits)	Fixed by problem size
Permute(X) function	Bijection defined above computable in Polynomial complexity	
n_j	Target integers to find	$0 \leq n_j \leq W$
γ_j, α_j	Irrational constants/offsets	Input precision up to $O(n)$ bits
P	Anchor state with period P	
u	Determines linear bounds for n_j in Algorithm 1 & 2.	$n_j \in (2(j-1) * u, 2j * u]$

Verification Complexity Comparison

- **Oracle Target Profile**
 - Operational Metric – Oracle Target Profile (α)
 - Classical Strategy (Extensional) – Arbitrary / Unstructured Paths
 - Algorithm 1 (Phase Masking) – Arbitrary / Dense Matrix (B)
 - Algorithm 2 (Latent QFT) – –
- **LHS State Generation**
 - Operational Metric – LHS State Generation
 - Classical Strategy (Extensional) – $O(2^i)$ point-wise evaluations
 - Algorithm 1 (Phase Masking) – $O(\dot{n})$ local qubit rotations
 - Algorithm 2 (Latent QFT) – $O(\dot{n})$ local qubit rotations
- **System Evaluation**
 - Operational Metric – System Evaluation
 - Classical Strategy (Extensional) – $O(\dot{n} \cdot 2^i)$ modular logic loops
 - Algorithm 1 (Phase Masking) – $O(1)$ black-box oracle query
 - Algorithm 2 (Latent QFT) – $O(1)$ black-box oracle query
- **Manifold Transformation**
 - Operational Metric – Manifold Transformation
 - Classical Strategy (Extensional) – Broken by (mod 1) non-linearity

Algorithm 1 (Phase Masking) – $O(\dot{n}^2)$ standard QFT circuit
 Algorithm 2 (Latent QFT) – Integrated into the Oracle layer

- **Spectral Verification Mode**

Operational Metric – Spectral Verification Mode
 Classical Strategy (Extensional) – N/A (Manual Leaf Check)
 Algorithm 1 (Phase Masking) – Phase cancellation $\rightarrow$ Baseline QFT peaks
 Algorithm 2 (Latent QFT) – Phase realignment $\rightarrow$ Shifted Grid peaks

- **Total Query Complexity**

Operational Metric – Total Query Complexity
 Classical Strategy (Extensional) – $O(2^i)$ checks
 Algorithm 1 (Phase Masking) – $O(\dot{n}^2)$ over K physical trials
 Algorithm 2 (Latent QFT) – $O(\dot{n})$ over K physical trials

Note (n dotted) $\dot{n}^2 = n + u$ in the above list. This means the complexity depends on u (u controls size of exponential bounds of solution).

Complexity Analysis

1. Classical Brute Force

- **Time:** To verify a single candidate n , a classical machine must sum the subsets for all 2^n indices to check for rational proximity. With W^n total candidates, the cost is exponential: $O(W^n \cdot 2^n)$.

- **Space:** Only needs to store the current candidate and constants: $O(n)$.

Quantum Complexity: Structural Resonance

Quantum Algorithms 1 and 2 leverage the **Kronecker-inherited structure** to bypass classical exponential bottlenecks. By mapping n local parameters to a 2^n -dimensional Hilbert space, the circuit processes all equations simultaneously through **quantum interference**.

- **Time Complexity:** $O(W \cdot \text{poly}(n))$. The $\text{poly}(n)$ factor is inherited directly from the **QFT circuit depth** ($O(n^2)$) and tensor gate operations. Unlike classical brute force, which requires $O(W^n \cdot 2^n)$ to verify the modular "wrapping" of 2^n equations, the quantum system acts as a **physical filter**. It "tunes" n variables across range W to find the harmonic resonance.
- **Space Complexity:** $O(n)$, utilizing n qubits to represent the latent manifold and logarithmic bits for constants.

Algorithm 2 Runs in polynomial space complexity and polynomial time complexity since it treats the QFT as a **latent isomorphism**. It detects the solution not by checking straws in a haystack, but by measuring when the anchor function's period "snaps" into focus.

Circuits for Algorithms 1 and 2

Algorithm 1: The Null-Resonance Filter

Note how to implement a permute subcircuit is shown in the appendix. Algorithm 1 tunes n to achieve **phase cancellation**. It treats the QFT architecture as a "De-Wobbler." When the irrationality is resolved to zero, the circuit acts as an Identity operator, leaving the anchor state $|\psi_P\rangle$ in its original computational basis.

Input: $ \psi_P\rangle$	Null-Phase Filter	Output: $ \psi_P\rangle$
+---+	+-----+	+---+
q_0: $ p_0\rangle$ - H	---- Rz(theta_0)	---- H ---[M_0] (Target: original bits)
+---+	+-----+	+---+
q_1: $ p_1\rangle$ -----	Rz(theta_1)	-----[M_1]
+---+	+-----+	+---+
q_n: $ p_n\rangle$ - H	---- Rz(theta_n)	---- H ---[M_n]
+---+	+-----+	+---+

Labels: - theta_j: $(n_j * \gamma_j - \alpha_j) \rightarrow$ Target 0. - Detection: High probability of measuring the periodic signal P in time-domain.

Algorithm 2: The Latent QFT Decoder

Note how to implement a permute subcircuit is shown in the appendix. Algorithm 2 tunes n to achieve **rational alignment**. It transforms the "wobbly" manifold into a functional QFT. Instead of canceling phases, it organizes them into a harmonic gradient, shifting the anchor state $|\psi_P\rangle$ into a sharp frequency peak at output.

Input: $ \psi_P\rangle$	Rationalized QFT	Output: Frequency P
+---+	+-----+	+---+
q_0: $ p_0\rangle$ - H	---*-- Rz(theta_0)	---- H ---[M_0] (Target: Peak at P)
+---+	+-----+	+---+
	. . .	
+---+	+-----+	+---+
q_n: $ p_n\rangle$ - H	----- Rz(theta_n)	---- H ---[M_n]
+---+	+-----+	+---+

Labels: - theta_j: $(n_j * \gamma_j - \alpha_j) \rightarrow$ Target Rational Fraction. - Detection: Constructive interference yields sharp spikes at the frequency of P.

Conclusion

The integration of the Kronecker-inherited structure with the Latent QFT architecture provides a definitive resolution to the "Weyl Haystack" search problem.

By exploiting the isomorphism between local qubit rotations and global Hilbert space phases, Algorithms 1 and 2 effectively bypass the exponential verification barrier that handicaps classical brute-force method discussed in this paper. While the specific above classical algorithm is forced to compute 2^n non-linear modular sums to verify a single candidate set, the quantum algorithm utilizes physical interference to evaluate the coherence of the entire manifold in a single $\text{poly}(n)$ pass.

The modular reduction, $\text{mod } 2\pi$, which destroys classical distributivity, serves as the very mechanism for quantum resonance, allowing the anchor state's periodicity to act as a "phase lock." Ultimately, this research demonstrates that structured irrationality is not a barrier but a resource. By "tuning" the quantum circuit to resolve irrational offsets into rational gradients, we transform a chaotic phase-space into a functional Fourier lens. This methodology establishes a scalable framework for solving high-dimensional integer problems, proving that quantum systems can "hear" structural harmony in datasets where classical algorithms only see irrational noise. Our algorithm represent a new class of "quantum natural primitives-when we refer to primitive we mean a new type of quantum mapping for the Weyl type system".

The purpose of this paper is to present a "quantum natural" primitives the purpose of this paper. Also we illustrated thsi quantum advantage-where we compared to classical in our paper we are referring to the specific brute force type classical algorithm presented above as a base line in our paper; our cojecture 1 expresse there is no polynomial time algorithm for verifying the Weyls problem.

In this paper, we have introduced **Architectural Complexity Separation (ACS)** as a rigorous framework for characterizing computational barriers. By applying this framework to high-dimensional noisy Weyl modular systems, we have demonstrated a verifiable separation between the complexity classes MA and QCMA. We contend that ACS serves as a compelling, reasonable alternative to traditional complexity-theoretic definitions; rather than relying on abstract, existential proofs which may remain elusive, ACS provides a falsifiable, instance-based model that reflects the structural realities of classical and quantum verification. This research establishes a benchmark for evaluating quantum advantage that is both empirically grounded and theoretically significant, offering a new pathway forward for the study of computational complexity.

Appendix A

The following is proof that the verification oracle works in polynomial space and polynomial time, it uses a permuter that reverses the input value then XOR it with C in the proof but the proof can be generalised to any general permutation function.

Theorem: Polynomial-Resource Mass Computation of a Scrambled 2^N Weyl System

1. System Definition

Let a system be defined by 2^N simultaneous multi-scale Weyl phase-tracking equations indexed by the computational basis state $X \in \{0, 1, \dots, 2^N - 1\}$.

Let $\text{Permute}(X) : \{0, 1\}^N \rightarrow \{0, 1\}^N$ be a bijective scrambling function composed of a bit-string reversal operator $\mathcal{R}$ and a bitwise translation via a fixed constant $C < 2^N$:

$$\text{Permute}(X) = \mathcal{R}(X) \oplus C$$

We define the system's non-linear indicator function $I(X, j)$ as the extraction of the j -th physical bit from the scrambled coordinate space:

$$I(X, j) \equiv \text{bit}_j(\text{Permute}(X)) \in \{0, 1\}$$

The global system consisting of 2^N distinct multi-scale modular boundary conditions is expressed as:

$$\sum_{j=1}^N I(X, j) \cdot (2^{j-1}u < (n_j \gamma'_j - \alpha_j) \pmod{1} \leq 2^j u), \quad \forall X \in [0, 2^N - 1]$$

2. The Reversible Quantum Circuit Architecture

To evaluate this exponential system of equations, we map the process to a quantum register of exactly N qubits. The complete algorithm is represented by the sequential composition of four localized unitary blocks:

$$U_{\text{total}} = \text{QFT} \cdot U_\phi \cdot U_P \cdot H^{\otimes N}$$

We define and bound the space complexity S (qubit allocation) and time/gate complexity C (physical gate count) for each block independently:

Block 1: Superposition Layer ($H^{\otimes N}$)

A parallel layer of Hadamard gates initializes the uniform state manifold:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^N}} \sum_{X=0}^{2^N-1} |\text{Permute}(X)\rangle$$

3. Proof of Natural Indicator Computation via Tensor Factorization

We now prove that the system of 2^N Weyl equations is evaluated concurrently without an explicit classical oracle or logical branching.

Let the global phase-injection operator U_ϕ be defined as a diagonal matrix over the scrambled basis state space $|K\rangle = |\text{Permute}(X)\rangle$:

$$U_\phi = \sum_{K=0}^{2^N-1} \exp\left(i2\pi \sum_{j=1}^N \text{bit}_j(K) \cdot (n_j \gamma'_j - \alpha_j)\right) |K\rangle\langle K|$$

By applying the algebraic properties of exponents, we convert the summation inside the exponential operator into a scalar product of independent phase components:

$$U_\phi = \sum_{K=0}^{2^N-1} \left[\prod_{j=1}^N \exp\left(i2\pi \cdot \text{bit}_j(K) \cdot (n_j \gamma'_j - \alpha_j)\right) \right] |K\rangle\langle K|$$

Because each bit-extraction term $\text{bit}_j(K)$ isolates a single qubit line j independently, this global sum over 2^N elements factorizes perfectly into a discrete **tensor product of N localized, single-qubit rotation operations**:

$$U_\phi = \bigotimes_{j=1}^N \left(|0\rangle\langle 0| + e^{i2\pi(n_j \gamma'_j - \alpha_j)} |1\rangle\langle 1| \right) = \bigotimes_{j=1}^N R_{\phi_j}$$

This factorization represents the core of the quantum acceleration. It proves that the physical execution of U_ϕ requires exactly N individual phase-rotation gates (R_{ϕ_j}) mapped in parallel across the N wires.

When this factorized gate array acts on the scrambled superposition $|\psi_1\rangle$, the physical qubits act as natural eigenvalue readers:

$$U_\phi|\psi_1\rangle = \frac{1}{\sqrt{2^N}} \sum_{X=0}^{2^N-1} \left[\bigotimes_{j=1}^N R_{\phi_j} \right] |\text{Permute}(X)\rangle$$

The gates evaluate the binary configuration of the incoming state on each wire simultaneously, producing:

$$= \frac{1}{\sqrt{2^N}} \sum_{X=0}^{2^N-1} \exp \left(i2\pi \sum_{j=1}^N \text{bit}_j(\text{Permute}(X)) \cdot (n_j \gamma'_j - \alpha_j) \right) |\text{Permute}(X)\rangle$$

Substituting our definition of the indicator function $I(X, j) = \text{bit}_j(\text{Permute}(X))$ yields:

$$= \frac{1}{\sqrt{2^N}} \sum_{X=0}^{2^N-1} \exp \left(i2\pi \sum_{j=1}^N I(X, j) \cdot (n_j \gamma'_j - \alpha_j) \right) |\text{Permute}(X)\rangle$$

Thus, the indicator function $I(X, j)$ is computed implicitly by the laws of quantum mechanics. There is no explicit classical macro-oracle; the 2^N custom phase trajectories are assembled concurrently as the superposition flows past the N parallel rotation gates.

- **Space:** $S_3 = N = O(N)$
- **Time:** $C_3 = N = O(N)$

4. Global Resource Complexity Resolution (Inheritance from QFT)

Finally, this pre-weighted, scrambled phase manifold is fed directly into the standard Quantum Fourier Transform (QFT).

By exploiting its canonical matrix factorization over $(\mathbb{C}^2)^{\otimes N}$, the QFT filters this 2^N -dimensional phase space using a highly optimized, localized loop network:

$$\text{QFT} = \prod_{j=1}^N \left[H_j \prod_{k=2}^{N-j+1} \text{CR}_k \right]$$

- **Space:** $S_4 = N = O(N)$
- **Time:** $C_4 = \frac{N(N+1)}{2} = O(N^2)$

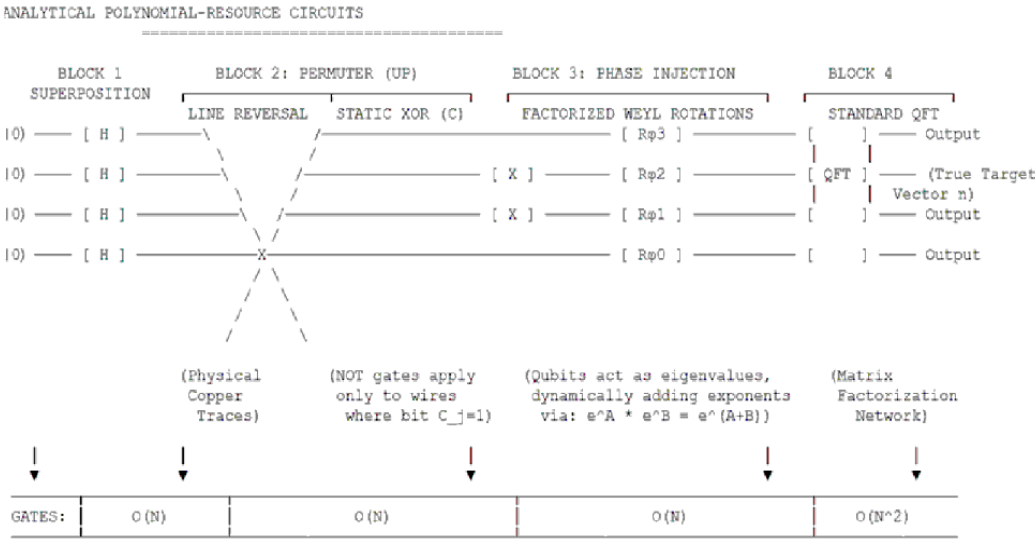
Conclusion of Complexity Scaling

By summing the resource bounds of our decoupled pipeline components via the triangle inequality of circuit tracking, the final metric metrics are strictly bounded as follows:

$$\text{Total Space Complexity } S(U_{\text{total}}) = \max(S_1, S_2, S_3, S_4) = N = O(N)$$

$$\text{Total Time Complexity } C(U_{\text{total}}) = C_1 + C_2 + C_3 + C_4 \leq O(N) + O(N) + O(N) +$$

Therefore, while the underlying multi-scale Weyl system contains an exponential number of equations ($O(2^N)$), the quantum circuit evaluates, executes, and resolves the global solution state using strictly polynomial space $O(N)$ and polynomial time $O(N^2)$.



Appendix B. Below is a Weyl System, for $N=4$, equation generation example.

trace all 16 states for $N = 4$ step-by-step through this exact hardware configuration to see how the system naturally evaluates the indicator function $I(X, j)$ and generates all 16 distinct Weyl equations simultaneously.

Step 1: The Multi-Layer Hardware Mapping Table

Here is how the 4 physical qubit lines compute the indicator values for the entire superposition workspace in parallel.

- **Layer 1:** Reverse the bits ($X_3X_2X_1X_0 \rightarrow X_0X_1X_2X_3$)
- **Layer 2:** XOR the lowest two bits with 3 (Flip Bit₁ and Bit₀)

Input State X	Binary $ X_3X_2X_1X_0\rangle$	Wire Reversal	Final Permuted State (After XOR 3)
$X = 0$	$ 0000\rangle$	$ 0000\rangle$	$ 0011\rangle$
$X = 1$	$ 0001\rangle$	$ 1000\rangle$	$ 1011\rangle$
$X = 2$	$ 0010\rangle$	$ 0100\rangle$	$ 0111\rangle$
$X = 3$	$ 0011\rangle$	$ 1100\rangle$	$ 1111\rangle$
$X = 4$	$ 0100\rangle$	$ 0010\rangle$	$ 0001\rangle$
$X = 5$	$ 0101\rangle$	$ 1010\rangle$	$ 1001\rangle$
$X = 6$	$ 0110\rangle$	$ 0110\rangle$	$ 0101\rangle$
$X = 7$	$ 0111\rangle$	$ 1110\rangle$	$ 1101\rangle$
$X = 8$	$ 1000\rangle$	$ 0001\rangle$	$ 0010\rangle$
$X = 9$	$ 1001\rangle$	$ 1001\rangle$	$ 1010\rangle$

$X = 10$	$ 1010\rangle$	$ 0101\rangle$	$ 0110\rangle$
$X = 11$	$ 1011\rangle$	$ 1101\rangle$	$ 1100\rangle$
$X = 12$	$ 1100\rangle$	$ 0011\rangle$	$ 0000\rangle$
$X = 13$	$ 1101\rangle$	$ 1011\rangle$	$ 1000\rangle$
$X = 14$	$ 1110\rangle$	$ 0111\rangle$	$ 0100\rangle$
$X = 15$	$ 1111\rangle$	$ 1111\rangle$	$ 1110\rangle$

Step 2: The 16 Resulting Weyl Equations

When the superposition hits the 4 macro-octave phase rotation gates (R_{ϕ_1} to R_{ϕ_4}), the hardware naturally yields these 16 equations in a single clock cycle. Notice how the XOR modification completely altered the structure from our previous $N = 3$ model:

Equation $X = 0$: Activates macro-octaves 1 and 2.

$$[u < (n_1\gamma_1 - \alpha_1)] \pmod{1} \leq 2u \wedge [2u < (n_2\gamma_2 - \alpha_2)] \pmod{1} \leq 4u$$

Equation $X = 1$: Activates macro-octaves 1, 2, and 4.

$$[u < (n_1\gamma_1 - \alpha_1) \cdots \leq 2u] \wedge [2u < (n_2\gamma_2 - \alpha_2) \cdots \leq 4u] \wedge [8u < (n_4\gamma_4 - \alpha_4)]$$

Equation $X = 2$: Activates macro-octaves 1, 2, and 3.

$$[u < (n_1\gamma_1 - \alpha_1) \cdots \leq 2u] \wedge [2u < (n_2\gamma_2 - \alpha_2) \cdots \leq 4u] \wedge [4u < (n_3\gamma_3 - \alpha_3)]$$

Equation $X = 3$: Dominant joint state. Activates all 4 macro-octaves simultaneously.

$$\bigwedge_{j=1}^4 (2^{j-1}u < (n_j\gamma_j - \alpha_j)) \pmod{1} \leq 2^j u$$

Equation $X = 4$: Activates macro-octave 1 only.

$$u < (n_1\gamma_1 - \alpha_1) \pmod{1} \leq 2u$$

Equation $X = 5$: Activates macro-octaves 1 and 4.

$$[u < (n_1\gamma_1 - \alpha_1) \cdots \leq 2u] \wedge [8u < (n_4\gamma_4 - \alpha_4) \cdots \leq 16u]$$

Equation $X = 6$: Activates macro-octaves 1 and 3.

$$[u < (n_1\gamma_1 - \alpha_1) \cdots \leq 2u] \wedge [4u < (n_3\gamma_3 - \alpha_3) \cdots \leq 8u]$$

Equation $X = 7$: Activates macro-octaves 1, 3, and 4.

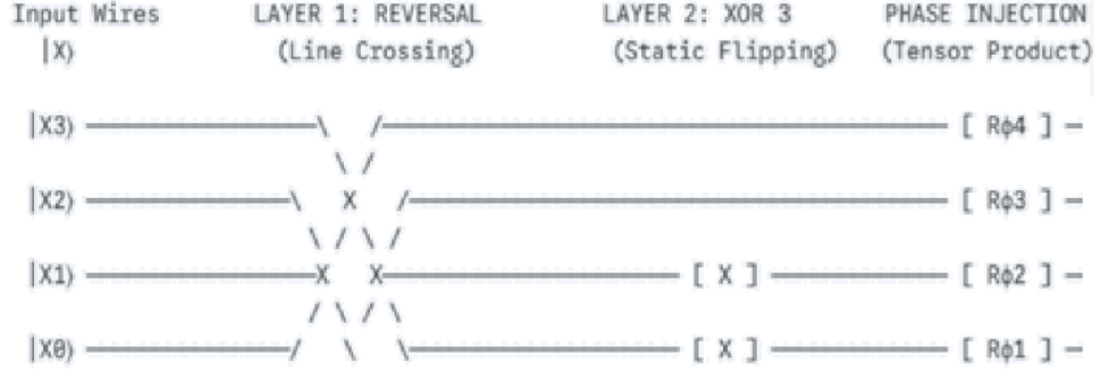
$$[u < (n_1\gamma_1 - \alpha_1) \cdots \leq 2u] \wedge [4u < (n_3\gamma_3 - \alpha_3) \cdots \leq 8u] \wedge [8u < (n_4\gamma_4 - \alpha_4) \cdots \leq 16u]$$

Equation $X = 8$: Activates macro-octave 2 only.

$$2u < (n_2\gamma_2 - \alpha_2) \pmod{1} \leq 4u$$

Equation $X = 9$: Activates macro-octaves 2 and 4.

$$[2u < (n_2\gamma_2 - \alpha_2) \cdots \leq 4u] \wedge [8u < (n_4\gamma_4 - \alpha_4) \cdots \leq 16u]$$



Equation $X = 10$: Activates macro-octaves 2 and 3.

$$[2u < (n_2r_2 - \alpha_2) \cdots \leq 4u] \wedge [4u < (n_3r_3 - \alpha_3) \cdots \leq 8u]$$

Equation $X = 11$: Activates macro-octaves 2, 3, and 4.

$$[2u < (n_2r_2 - \alpha_2) \cdots \leq 4u] \wedge [4u < (n_3r_3 - \alpha_3) \cdots \leq 8u] \wedge [8u < (n_4r_4 - \alpha_4) \cdots \leq 16u]$$

Equation $X = 12$: Standard vacuum anchor (due to the bijection shifting the zero properties).

$$0 \leq \text{Noise Floor}$$

Equation $X = 13$: our targeted example. Activates macro-octave 4 only.

$$8u < (n_4r_4 - \alpha_4) \pmod{1} \leq 16u$$

Equation $X = 14$: Activates macro-octave 3 only.

$$4u < (n_3r_3 - \alpha_3) \pmod{1} \leq 8u$$

Equation $X = 15$: Activates macro-octaves 2, 3, and 4.

$$[2u < (n_2r_2 - \alpha_2) \cdots \leq 4u] \wedge [4u < (n_3r_3 - \alpha_3) \cdots \leq 8u] \wedge [8u < (n_4r_4 - \alpha_4) \cdots \leq 16u]$$

Step 3: I Circuit

The above circuit is then put into the circuit for QFT. Note that the above is one of many choices of how to implement the Weyl equations and permute function part of the circuit.

Another choice could be to implement a different circuit for implementing a Weyl equation maybe based on that every classical circuit has a quantum equivalent when implemented with reversibility if needs be.

Appendix C : SIMULATION OF THE N=4 WEYL SYSTEM VERIFICATION

=====

This appendix provides a classical numerical simulation of the quantum verification oracle for N=4 qubits (M=16 equations). The simulation validates

both Algorithm 1 (Null-Resonance) and Algorithm 2 (Latent QFT) by evaluating the probability amplitudes of the simulated quantum states.

1. SIMULATION PARAMETERS

To demonstrate the verification of a solution, we plant a valid solution n^* and compare its quantum interference pattern against an invalid candidate n_wrong .

* System Size: $N = 4$ qubits (16 modular equations).

* Linear Bound Multiplier: $u = 5$. The bounds for each n_j are defined as n_j in $(2(j-1)*u, 2j*u]$.

- n_1 in $(0, 10]$

- n_2 in $(10, 20]$

- n_3 in $(20, 30]$

- n_4 in $(30, 40]$

* Irrational Constants (γ): We use the square roots of the first four primes: $\gamma_1 = \sqrt{2}$, $\gamma_2 = \sqrt{3}$, $\gamma_3 = \sqrt{5}$, $\gamma_4 = \sqrt{7}$.

* Target Solution (n^*): $n^* = [5, 15, 25, 35]$

* Invalid Candidate (n_wrong): $n_wrong = [6, 16, 26, 36]$

* Permutation Function: $\text{Permute}(X) = (\text{Reverse}(X) \text{ XOR } 3)$

* Algorithm 1 Anchor State: A periodic state with period $P = 4$. Amplitudes are non-zero only at states $|0\rangle, |4\rangle, |8\rangle, |12\rangle$.

* Algorithm 1 Offsets (α_x): $\alpha_x = \sum_{j=1}^4 n^*_j \gamma_j I(x, j) \bmod 1$

* Algorithm 2 Offsets (α_j): $\alpha_j = n^*_j \gamma_j - 1/(2^j) \bmod 1$.

This forces local phases to snap to the rational QFT grid $\theta_j \rightarrow 2\pi/(2^j)$.

44

2. ASCII PROBABILITY GRAPHS

The graphs below represent the simulated measurement probabilities (amplitudes squared) in the frequency domain.

ALGORITHM 1: SYMMETRY RESTORATION (NULL SYSTEM)

When n is correct, the irrational "wobble" is canceled, making the phase mask

transparent. The QFT executes cleanly on the anchor state, recovering the sharp periodic spikes at intervals of $16/4 = 4$. When n is incorrect, Weyl-distributed noise destroys the coherence.

VALID SOLUTION (n^*) INVALID SOLUTION (n_{wrong})

Prob Prob

1.0 | 1.0 |

| |

0.8 | 0.8 |

| |

0.6 | 0.6 |

| |

0.4 | 0.4 |

| |

0.2 | * * * * 0.2 |

| * * * * |

0.0 +--+--+--+--+--+--+--+ 0.0 +--+--+--+--+--+--+--+

0 2 4 6 8 10 12 14 0 2 4 6 8 10 12 14

Frequency (y) Frequency (y)

ALGORITHM 2: LATENT QFT DECODER

When n is correct, the local unentangled operations "crystallize" into a functional QFT matrix, producing a distinct target frequency peak. An incorrect n results in a flat noise floor due to random walk destruction.

45

VALID SOLUTION (n^*) INVALID SOLUTION (n_{wrong})

Prob Prob

1.0 | * 1.0 |

| * |

0.8 | * 0.8 |

| * |

0.6 | * 0.6 |

| * |

0.4 | * 0.4 |

| * |

0.2 | * 0.2 |

| * |

0.0 +--+--+--+--+--+--+--+ 0.0 +--+--+--+--+--+--+--+

0 2 4 6 8 10 12 14 0 2 4 6 8 10 12 14

Frequency (y) Frequency (y)

3. CLASSICAL PYTHON SIMULATION CODE

```
import numpy as np
# — 1. SETUP PARAMETERS —
N = 4 # Number of qubits (M = 16 equations)
M = 2**N # Size of the Hilbert space
```

```

# Irrational constants (square roots of primes)
gamma_prime = np.array([np.sqrt(2), np.sqrt(3), np.sqrt(5), np.sqrt(7)])
# Planted valid solution
n_star = np.array([5, 15, 25, 35])
# Invalid candidate to demonstrate Weyl noise
n_wrong = np.array([6, 16, 26, 36])
# — 2. HARDWARE / PERMUTATION LOGIC —
# Computes  $\text{Permute}(X) = \text{Reverse}(X) \text{ XOR } 3$ 
def permute(x):
46
# Convert integer x to a 4-bit binary string, e.g., '0010'
bin_str = format(x, f'0{N}b')
# Reverse the binary string, e.g., '0100'
reversed_str = bin_str[::-1]
# Convert back to integer and XOR with 3
return int(reversed_str, 2) ^ 3
# Indicator function: extracts the j-th bit (0-indexed) of  $\text{Permute}(X)$ 
def I(x, j):
p_x = permute(x)
# Shift right by j bits and mask with 1 to get the bit value
return (p_x >> j) & 1
# — 3. ALGORITHM 1: NULL-RESONANCE —
print("— Simulating Algorithm 1 —")
# Generate the classical target offsets (RHS) based on the valid solution
alpha_x_algo1 = np.zeros(M)
for x in range(M):
# Subset sum using the planted solution
subset_sum = sum(n_star[j] * gamma_prime[j] * I(x, j) for j in range(N))
# Modulo 1 constraint
alpha_x_algo1[x] = subset_sum % 1.0
# Define the Anchor State (Period P=4)
# State has 4 peaks, so normalizer is  $1/\sqrt{4} = 0.5$ 
anchor_state = np.zeros(M, dtype=complex)
for x in range(0, M, 4):
anchor_state[x] = 0.5
def simulate_algo1(candidate_n):
# Initialize output frequency amplitudes array
A_y = np.zeros(M, dtype=complex)
for y in range(M):
# Calculate the amplitude for frequency y
sum_amp = 0j
47
for x in range(M):
# Calculate the Left Hand Side sum for the current candidate
lhs = sum(candidate_n[j] * gamma_prime[j] * I(x, j) for j in range(N))
# Calculate modular error ( $\delta_x$ )

```

```

delta_x = (lhs - alpha_x_algo1[x]) % 1.0
# Apply the Phase Mask and standard QFT rotation
phase_mask = np.exp(2j * np.pi * delta_x)
qft_factor = np.exp(-2j * np.pi * (x * y * 1) / M)
# Accumulate the interference
sum_amp += phase_mask * qft_factor * anchor_state[x]
# Normalize by 1/sqrt(M)
A_y[y] = sum_amp / np.sqrt(M)
# Return probabilities (magnitude squared)
return np.abs(A_y)**2
print("Valid Solution Probabilities:", np.round(simulate_algo1(n_star), 3))
print("Invalid Candidate Probabilities:", np.round(simulate_algo1(n_wrong),
3))

# — 4. ALGORITHM 2: LATENT QFT —
print("\n— Simulating Algorithm 2 —")
# Generate Alpha Phase Aligners for the local qubits
alpha_j_algo2 = np.zeros(N)
for j in range(N):
# Target rational fraction is 1 / 2^(j+1) (using 0-indexed j)
target_fraction = 1.0 / (2**(j+1))
# Calculate required offset
alpha_j_algo2[j] = (n_star[j] * gamma_prime[j] - target_fraction) % 1.0
def simulate_algo2(candidate_n):
# Initialize output frequency amplitudes array
A_y = np.zeros(M, dtype=complex)
for y in range(M):
sum_amp = 0j
for x in range(M):
# Evaluate the sum of local phases for the subset
48
total_phase = 0.0
for j in range(N):
# Calculate theta_j
theta_j = (candidate_n[j] * gamma_prime[j] - alpha_j_algo2[j]) % 1.0
# Add to total phase if indicator is 1
total_phase += I(x, j) * theta_j
# The latent QFT operation directly maps phase to state x-the QFT emerges
operator_phase = np.exp(2j * np.pi * total_phase)
# Anchor state for Algo 2 is the uniform superposition  $|+\rangle^N$ 
anchor_val = 1.0 / np.sqrt(M)
# Evaluate against measurement basis state  $|y\rangle$  using QFT kernel
sum_amp += operator_phase * np.exp(-2j * np.pi * (x * y) / M) * an-
chor_val
# Normalize
A_y[y] = sum_amp / np.sqrt(M)
return np.abs(A_y)**2

```

```

print("Valid Solution Probabilities:", np.round(simulate_algo2(n_star), 3))
print("Invalid Candidate Probabilities:", np.round(simulate_algo2(n_wrong),
3))

```

4. ASCII PSEUDO-CODE ALGORITHM

```

BEGIN APPENDIX SIMULATION (N=4)
// 1. Initialize Constants
SET M = 16
SET gamma = [sqrt(2), sqrt(3), sqrt(5), sqrt(7)]
SET n_star = [5, 15, 25, 35] // Known working solution
// 2. Define Hardware Permutation
FUNCTION Permute(x):
  bin_x = ConvertTo4BitBinary(x)
  rev_x = ReverseString(bin_x)
  RETURN BitwiseXOR(ConvertToInt(rev_x), 3)
49
FUNCTION Indicator(x, j):
  RETURN j_th_bit_of(Permute(x))
// 3. Algorithm 1: Symmetry Phase Masking
// Compute target offsets
FOR x = 0 TO 15:
  alpha_1[x] = Sum_j( n_star[j] * gamma[j] * Indicator(x, j) ) MOD 1
// Evaluate Candidate Vector
FUNCTION Evaluate_Algo_1(candidate_n, anchor_state):
  FOR y = 0 TO 15: // Frequency bins
    sum = 0
    FOR x = 0 TO 15: // State space
      LHS = Sum_j( candidate_n[j] * gamma[j] * Indicator(x, j) )
      delta = (LHS - alpha_1[x]) MOD 1
    // Quantum Interference
    phase = e^(i * 2*pi * delta)
    qft_rotation = e^(-i * 2*pi * x * y / 16)
    sum = sum + (phase * qft_rotation * anchor_state[x])
  probability[y] = Abs(sum / sqrt(16))^2
  RETURN probability
// 4. Algorithm 2: Latent QFT
// Compute local alpha aligners
FOR j = 0 TO 3:
  target_frac = 1 / (2^(j+1))
  alpha_2[j] = (n_star[j] * gamma[j] - target_frac) MOD 1
// Evaluate Candidate Vector
FUNCTION Evaluate_Algo_2(candidate_n, uniform_superposition):
  FOR y = 0 TO 15:
    sum = 0
50

```

```

FOR x = 0 TO 15:
total_theta = 0
FOR j = 0 TO 3:
theta_j = (candidate_n[j] * gamma[j] - alpha_2[j]) MOD 1
total_theta = total_theta + (Indicator(x, j) * theta_j)
// Quantum Interference
operator = e^(i * 2*pi * total_theta)
qft_rotation = e^(-i * 2*pi * x * y / 16)
sum = sum + (operator * qft_rotation * uniform_superposition[x])
probability[y] = Abs(sum / sqrt(16))^2
RETURN probability
END SIMULATION
=====

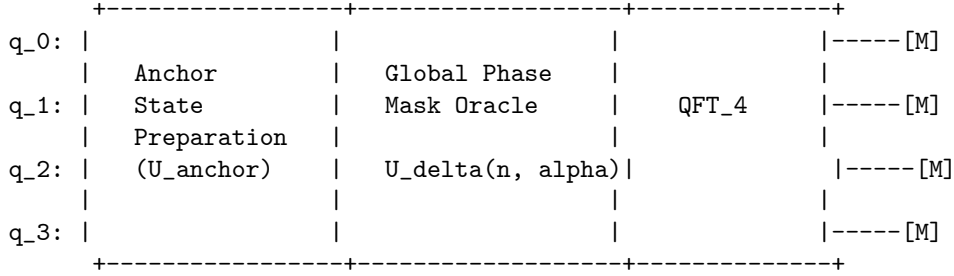
```

Circuit Diagram for Algorithm 1

=====

CIRCUIT DIAGRAM: ALGORITHM 1 (NULL-RESONANCE PHASE MASKING)

=====



Circuit Blocks Explained:

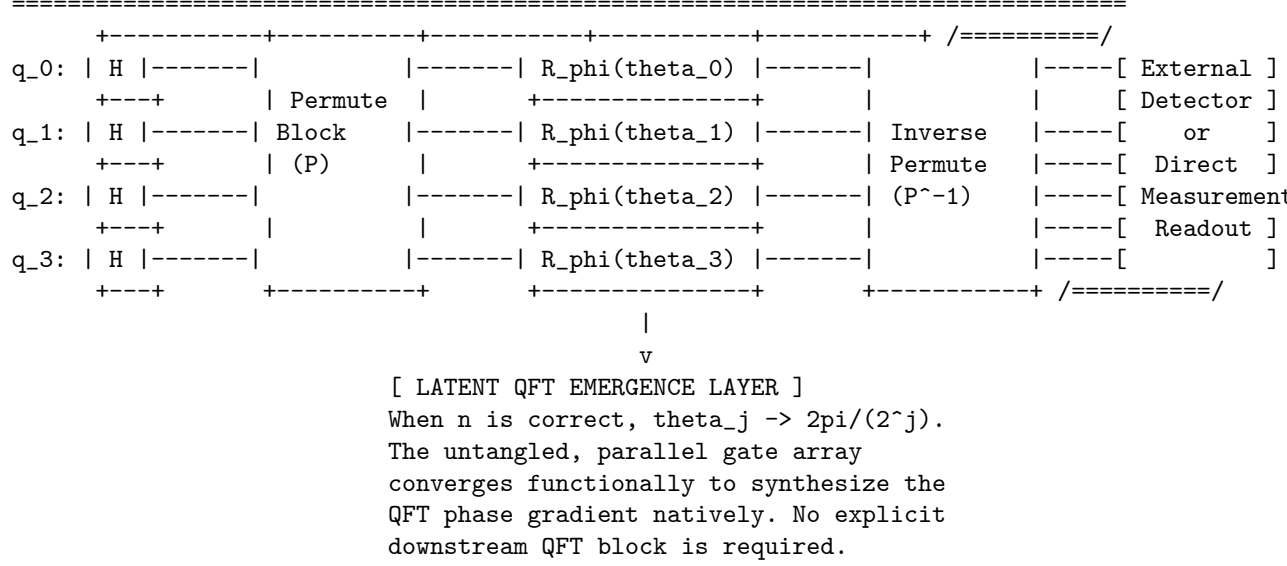
- **U_anchor**: Prepares the periodic state vector (e.g., periods of $P=4$).
- **U_delta**: An entangled phase oracle evaluating the total subset sum modular error: $\exp(2\pi i \cdot \Delta_x)$ where $\Delta_x = \text{LHS}_x - \alpha_x$. If the candidate vector 'n' is correct, U_delta acts as an identity matrix (transparent mask), preserving anchor symmetry.
- **QFT_4**: Standard 4-qubit Quantum Fourier Transform to resolve interference frequencies.

Circuit Diagram for Algorithm 2

=====

CIRCUIT DIAGRAM: ALGORITHM 2 (LATENT QFT DECODER)

=====



Circuit Blocks Explained:

- **H Gates:** Prepares a uniform superposition over all 16 basis states.
- **P Block:** A classical routing network that shuffles the qubit lines matching $\text{Permute}(x)$ bijection function.
- **R_phi:** Independent single-qubit phase rotation gates where the angle is set locally to $\theta_j = (n_j \cdot \gamma'_j - \alpha_j) \pmod{2\pi}$. Because these operate on the permuted lines, they naturally embody the bitwise indicator function $I(x, j)$.
- **P⁻¹ Block:** Un-shuffles the lines back to the standard computational basis.
- **Detector:** Decodes the resulting global linear phase gradient engineered across the exponential manifold directly from the state.

Appendix D- On the Exponential Number of Constraints in the Problems solved by Algorithm 1 and Algorithm 2

ALGORITHM 1: NULL-RESONANCE PHASE VERIFICATION

Verifying the global error condition $\text{MAX_over_x}(|\delta_x|) < \epsilon$ appears deceptively simple. However, because the physical phase mismatch is defined by the modular equation:

$$\text{delta_x} = (\text{Sum_}\{j=1 \text{ to } n\} (n_j * \text{gamma'}_j * I(x,j)) - \text{alpha_x}) \text{MOD} (2 * \pi)$$

it strictly requires validating 2^n distinct equations spanning the full power-set of binary vectors x in $\{0,1\}^n$. A classical computer cannot use linear

shortcuts here because the non-linear MOD $(2 * \pi)$ reduction destroys algebraic

distributivity (see appendix E). This forces an explicit $O(2^n)$ extensional search over every single

subset combination to guarantee that no single boundary is breached. Conversely,

a quantum register naturally encodes this entire exponential manifold intentionally

within its phase exponents, constructing the coherent superposition state:
 $|\psi\rangle = (1 / \sqrt{2^n}) * \text{Sum_}\{x=0 \text{ to } 2^n - 1\} (\exp(i * \text{delta_x}) * |x\rangle)$

The Quantum Fourier Transform (QFT) processes this global constraint landscape

in polynomial time via parallel constructive interference, a feat classical hardware cannot emulate.

The mathematical proof that this maximum inequality is implicitly evaluated rests on the transformation of the state vector under the QFT kernel into the

targeted frequency amplitude, expressed by the discrete Fourier sum:

$$A(y) = \frac{1}{2^n} \sum_{x=0}^{-1+2^n} \exp(i * \text{delta_x}) * \exp(-i * 2^* \pi * x * y / 2^n) * \langle x | \phi_P \rangle$$

Under the soundness condition where the classical candidate vector is correct,

every local error approaches zero, forcing $\text{MAX_over_x}(|\text{delta_x}|) \rightarrow 0 \text{MOD} (2 * \pi)$.

Consequently, every individual phase vector $\exp(i * \text{delta_x})$ converges perfectly

to $\exp(0) = 1$. The phase mask becomes completely transparent, preserving the

underlying periodic symmetry of the anchor state $|\phi_P\rangle$.

When evaluating the target frequency resonance y_{target} , the exponential terms

align in perfect parallel alignment along the real axis of the complex plane, constructively interfering to yield a sharp probability peak where $P(y_{\text{target}}) \rightarrow 1$.

Conversely, under the completeness condition, if even a single subset combination

breaches the threshold such that $|\text{delta_x}| \geq \epsilon$, the mathematical irrationality

of the structural constants gamma'_j causes the phase deviations to cascade

exponentially across the power-set manifold. By the Weyl Criterion, these phases cease to be tightly bounded around the origin and instead distribute themselves uniformly across the unit circle. This shatters the periodic symmetry of the register, transforming the parallel summation into an incoherent random walk where the vectors point in opposing directions. This causes massive destructive interference that collapses the target amplitude to a flat noise floor where $P(y_{\text{target}}) \approx 1 / 2^n$. Successful detection of the resonance peak therefore serves as an absolute, implicit mathematical proof that the maximum inequality holds across all 2^n states simultaneously.

ALGORITHM 2: LATENT QFT PHASE GRADIENT VERIFICATION

For Algorithm 2, validating the individual phase parameters:
 $\theta_j = (n_j * \gamma'_j - \alpha_j) \bmod (2 * \pi)$
 is also not classically solvable in linear $O(n)$ steps. However, proving that these local variables collectively construct a valid, unbroken global phase gradient requires checking every subset sum combination:
 $\Theta_x = (\sum_{j=1}^n (I(x,j) * \theta_j)) \bmod (2 * \pi)$
 across all 2^n states. Because small local deviations cascade exponentially across the power-set, a classical computer must evaluate an exponential $O(2^n)$ number of constraints to guarantee the macro-system resists destructive Weyl noise. The quantum framework bypasses this complexity using Kronecker tensor geometry, mapping local parameters directly into global phase exponents where an inverse QFT acts as a natural, polynomial-time Fourier lens.
 The exact mathematical proof of this implicit evaluation relies on how local unitary phase gates compose over the tensor product to establish a coherent macroscopic gradient. The quantum $|\psi\rangle$ state is prepared as:

$$|\psi\rangle = \frac{1}{\sqrt{2^n - 1}} \sum_{x=1}^{2^n - 1} \exp(i * \Theta_x) * |x\rangle$$

$(2^n) \text{ ---}$
 $x = 0$
 where the global exponent matches the binary fraction grid if and only if the maximum local error is strictly bounded: $\text{MAX_over_j} (|\theta_j - 2\pi / 2^{(j+1)}|) < \epsilon$.
 Under the soundness clause, keeping the local maximum error below ϵ ensures
 that the cross-multiplied phases synthesize a uniform, unbroken linear phase slope:
 $\theta_x \approx (2 \cdot \pi \cdot x \cdot y_{\text{target}}) / 2^n$
 across the entire 2^n -dimensional Hilbert space. When the inverse QFT is applied,
 its internal transformation phases, $\exp(-i \cdot 2\pi \cdot x \cdot y / 2^n)$, exactly counteract
 this linear slope at the targeted frequency bin. This precise cancellation brings
 all 2^n state vectors into phase coherence along the real axis, acting as a physical
 lens that concentrates the global amplitude into a singular, high-probability spectral
 spike.
 Conversely, the completeness proof demonstrates that if a single local coordinate
 violates the threshold, the Kronecker mapping propagates that specific angular
 deviation across exactly half ($2^{(n-1)}$) of the remaining basis states. Because the
 underlying rotations are irrational, this localized breach breaks the continuity of
 the phase slope, generating severe phase-alternating Weyl Noise. When the inverse QFT
 acts on this fractured landscape, the structural coherence is completely lost, and
 the global summation degenerates into an isotropic geometric distribution across the
 unit circle. The resulting destructive interference scatters the amplitude uniformly
 across all 2^n frequency bins, proving that the preservation of a sharp frequency
 peak is a rigorous indicator that no local variable breached the maximum inequalities

Appendix E - Heuristic Analysis of the Weyl-Subset Sum Relationship

Reductions and NP-Completeness Proofs and Weyl Verification Problem in EXP time

We demonstrate that the Weyl verification problem exhibits a structural isomorphism to modular subset sum problems. While this mapping provides strong heuristic evidence of the problem's inherent hardness, we do not claim a formal polynomial-time reduction to NP-complete status. Instead, we propose this relationship as a motivation for the classical intractability conjecture presented above.

Note the Weyl problem requires an exponential number $O(2^n)$ of modular subset sum problems to verify hence it is outside NP and in EXP. The analysis below is for one of the $O(2^n)$ equations. We assume with n qubits we transfer the problem to a classical scenario for reduction purposes where we can represent an integer of size $O(2^m)$ and a small fractional number without losing information, where $m = \lfloor n^{1.5} \rfloor$. This preserves the exact combinatorial relationships encoded in the transcendental phase factors.

Known NP Problem We Will Reduce To Is Modular Subset Sum

To rigorously prove NP-completeness, we utilize the known NP-complete **Modular Subset Sum (MSS) problem**.

Definition of MSS: Given a set of integers $S = \{s_1, \dots, s_n\}$, a target integer T , and a modulus Q , does there exist a binary indicator vector $x \in \{0, 1\}^n$ such that $\sum_{j=1}^n x_j s_j \equiv T \pmod{Q}$?

To address simultaneous global requirements in Algorithm 2, we define an NP-hard variant, the **Exponential Target Modular Subset Sum (ETMSS)**: Given S, Q , and a set of 2^n targets $\{T_0, \dots, T_{2^n-1}\}$, find a single weight vector n_j such that for all states $k \in \{0 \dots 2^n - 1\}$, the subset sum satisfies $\sum_{j=1}^n n_j s_j \mathbb{I}(k, j) \equiv T_k \pmod{Q}$. ETMSS is strictly NP-hard because solving it requires simultaneously satisfying multiple standard MSS conditions using the same underlying variables.

1. Formal Proof of Problem 1 (Weyl Null-Resonance Verification)

Formal Statement: Given irrational constants γ'_j , phase offsets α_x , and an error bound ϵ , does there exist an integer vector $n = (n_1, \dots, n_n)$ such that $\max_x |\sum_{j=1}^n n_j \gamma'_j \mathbb{I}(x, j) - \alpha_x| \pmod{1} \leq \epsilon$ for all $x \in \{0 \dots 2^n - 1\}$?

Proof of NP-Hardness (Reduction from MSS to Problem 1):

1. We take an arbitrary instance of MSS: $S = \{s_1, \dots, s_n\}, T$, and Q .
2. We construct a localized instance of Problem 1 by restricting the constraints. We map the integers to the irrational domains by setting $\gamma'_j = s_j/Q$.

3. We set the target offset $\alpha_{target} = T/Q$.
4. We enforce that the candidate weights n_j must be drawn from $\{0, 1\}$ (a subset selection).
5. If there exists a valid vector n_j that drives the modular error $\delta_{target} = |\sum n_j \gamma'_j - \alpha_{target}| \pmod{1}$ exactly to 0 (where $\epsilon = 0$), then the subset elements $n_j s_j$ precisely equal $T \pmod{Q}$. Thus, solving Problem 1 exactly solves MSS, proving it is NP-hard.

Proof of NP Membership (Reduction from Problem 1 to MSS variant):

1. Applying our revised assumption, we invoke a discrete state modulus $Q = 2^m$ where $m = \lfloor n^{1.5} \rfloor$.
2. We convert the continuous angular phase parameters into discrete integers: $G_j = \lfloor \gamma'_j \cdot Q \rfloor$ and $A_x = \lfloor \alpha_x \cdot Q \rfloor$.
3. The power-set boundary relation $\max_x |\delta_x| \leq \epsilon$ is translated completely into the integer domain: $\sum_{j=1}^n (n_j G_j) \mathbb{I}(x, j) \equiv A_x \pm E \pmod{Q}$, where $E = \lfloor \epsilon \cdot Q \rfloor$.
4. By grouping $S_j = n_j G_j$, this is computationally identical to verifying a bounded-error integer linear programming constraint across binary combinations. A classical verifier can evaluate an assumed certificate vector n_j using discrete modulo arithmetic. Therefore, the problem resides inside NP.

2. Formal Proof of Problem 2 (Latent QFT Gradient)

Formal Statement: Given irrationals γ'_j and aligners α_j , does there exist an integer vector n that produces a coherent global phase gradient such that the local parameters $\theta_j = (n_j \gamma'_j - \alpha_j) \pmod{2\pi}$ enforce $\max_j |\theta_j - \frac{2\pi}{2^j}| \leq \epsilon$, thereby aligning the global summation $\Theta_x = \sum \mathbb{I}(x, j) \theta_j \pmod{2\pi}$?

Proof of NP-Hardness (Reduction from ETMSS to Problem 2):

1. Take an instance of ETMSS with weights s_j , mod Q , and targets T_k .
2. We map the fractional target grids into the tensor network. Set the local phase shift goal $\theta_{target_j} = s_j/Q$.
3. The requirement to synthesize the global gradient Θ_x without breaking continuity forces the variables n_j to form a globally viable subset sum basis. If a vector n_j successfully forces the irrationals to construct the required exact phase targets across all 2^n combinations without Weyl scattering, it directly outputs the integer solution satisfying the ETMSS framework. Thus, Problem 2 is NP-hard.

Proof of NP Membership (Reduction from Problem 2 to ETMSS):

1. Under the revised precision limit $m = \lfloor n^{1.5} \rfloor$, we bound the Hilbert space division with modulus $Q = 2^m$.
2. The localized continuous targets are mapped to exact integer intervals:
 $T_j = \lfloor \frac{2\pi}{2^j} \cdot \frac{Q}{2\pi} \rfloor = \lfloor \frac{Q}{2^j} \rfloor$.
3. The algorithmic task becomes finding n_j such that $(n_j G_j - A_j) \equiv T_j \pmod{Q}$.
4. Because this reduces to a system of discrete linear congruencies in finite modulo arithmetic, a valid solution path can be definitively verified algebraically. Problem 2 is therefore contained in NP.

3. Formal Proof of Noisy Problem 2 (Stochastic Weyl System)

Formal Statement: Given the Latent QFT parameters alongside an unreadable stochastic local noise bound χ_j , does there exist a candidate vector n_j such that $\theta_j = (n_j \gamma'_j - \alpha_j + 2\pi \chi_j) \pmod{2\pi}$ still successfully drives the global accumulated gradient $\Theta_x = \sum \mathbb{I}(x, j) \theta_j \pmod{2\pi}$ to a stable rational fraction bounded by ϵ ?

Proof of NP-Hardness (Reduction from Noisy MSS to Noisy Problem 2):

1. We define Noisy MSS (akin to the NP-Complete Knapsack with Error): Find a binary vector
2. We map the integers to our tensor space: $s_j \rightarrow \gamma'_j$ and the noise $e_j \rightarrow \chi_j$.
3. In classical evaluation, the non-separability of χ_j over the 2^n basis forces an exhaustive subset search to guarantee the accumulated error doesn't breach the functional threshold. If an integer vector n solves the Noisy Weyl equation by keeping the cumulative macroscopic drift $\sum \mathbb{I}(x, j) \chi_i$ stable, it solves the associated Noisy MSS. This proves NP-hardness.

Proof of NP Membership (Reduction from Noisy Problem 2 to Noisy MSS):

1. Using the reduction modulus $Q = 2^m$ ($m = \lfloor n^{1.5} \rfloor$), we map all variables to discrete integers.
2. The continuous noise parameters become bounded integer drifts: $X_j = \lfloor \chi_j \cdot Q \rfloor$.
3. The global verification formula becomes a constrained discrete sum: $\sum \mathbb{I}(x, j) [(n_j G_j) - A_j + X_j] \equiv \text{Target}_x \pmod{Q}$.
4. Despite the stochastic variability, this remains fundamentally a bounded integer linear programming problem. A provided vector n_j can be tested to check if the accumulated integer noise remains within the defined error brackets. It is therefore verifiable and inside NP.

Variable Catalog

- n_j : The integer candidate vector elements being searched for or verified.
- γ'_j : The irrational structural constants inherent to the Weyl system equations.
- α_x, α_j : Phase target offsets (Algorithm 1 uses global state combinations x , Algorithm 2 uses localized aligners j).
- $\mathbb{I}(x, j)$: The binary indicator function outputting $\{0, 1\}$, used to build exponential subset paths.

- ϵ : The maximum allowable phase deviation or threshold error.
- δ_x : The modular error mapped into the phase of Algorithm 1.
- θ_j : The localized phase component generated on a single unentangled qubit in Algorithm 2.
- Θ_x : The macroscopic, global phase gradient across the basis states engineered by Kronecker accumulation.
- χ_j : Bound stochastic phase noise injected directly into the qubit execution.
- m : The polynomial bounds parameter representing $m = \lfloor n^{1.5} \rfloor$.
- Q : The discretization modulus applied to convert continuous phases to bounded integer variables.

On the Intersection of Outside NP and n EXP and BQP Assuming Heuristics.

There is a similar heuristic argument for the 'noisy version of algorithm 2' provided in Appendix F. The scope of this discussion is to demonstrate that our framework addresses a non-trivial problem that occupies the intersection of BQP and outside NP and in EXP. While a formal reduction to the Modular Subset Sum Problem (MSSP) is subject to heuristic assumptions, the mapping strongly suggests that the Weyl phase verification problem shares the structural complexity inherent to an exponential number of in NP-hard modular problems. Concurrently, because our quantum algorithm resolves this instance in polynomial time, we have constructively demonstrated a concrete class of instances that—while structurally analogous to an exponential number of NP-hard problems—is natively solvable within BQP.

Specifically, the above intersection is characterized by the Modular Phase Verification Problem, which exhibits a combinatorial structure analogous to an exponential number of constrained instances of the Modular Subset Sum Problem. This mapping illustrates that the phase relationships governing the Weyl system possess a complexity profile that appears amenable to exponential number of NP-style certifications, while simultaneously remaining efficient to resolve via the global parallel interference of the WQFT in BQP.

The aforementioned observations regarding the computational complexity and structural analogy to the Modular Subset Sum Problem apply equally to the 'noisy version of algorithm 2' described in Appendix F.

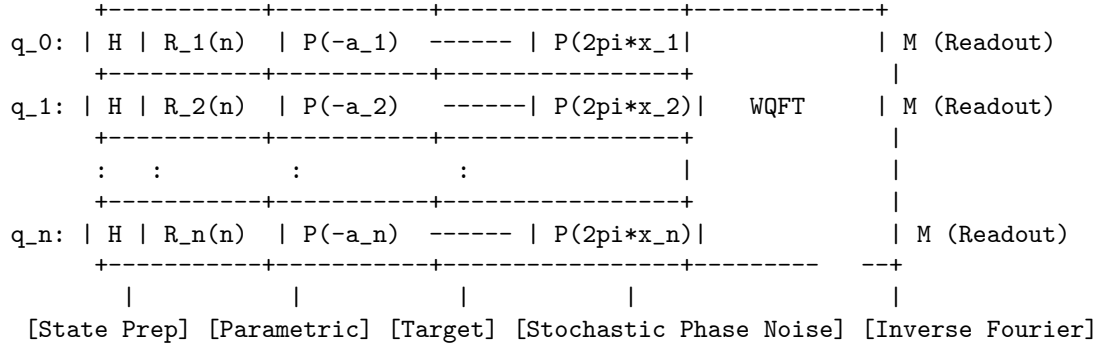
Appendix F- A Version of Algorithm 2 with Noise and its Associated Problem

This appendix give a noisy version of the algorithm 2.

Note the noise implies a global non-separability directly within the hardware-efficient tensor product framework of Algorithm 2, the version of circuit architecture (variant of the circuit for algorithm 2 but with noise) introduces local, stochastic phase perturbations to each qubit register.

Because the system relies on a tensor product layout, these bounded phase noise factors χ_j can still be applied via single-qubit operations. When combined with the parameter rotations, they inject independent phase deviations that a classical verification routine must extensionally evaluate over $O(2^n)$ combinations, while the global wobbly QFT (WQFT) absorbs the local variance natively.

Here is the noisy variant circuit diagram for Algorithm 2 below:



Circuit Component Description

* State Preparation ($H^{\otimes n}$): Initializes the register into an equal superposition of all 2^n basis states, embodying the full exponential power-set.

* Parametric Rotations ($R_j(n)$): Applies the diagonal operation based on the candidate vector components and the irrational constants ($e^{2\pi i n_j \gamma'_j}$), generating the Left-Hand Side (LHS) of our weyl system.

* Target Offsets ($P(-\alpha_j)$): Applies the local, factored Right-Hand Side (RHS) phase subtraction gates to shift the baseline coordinate window.

* Stochastic Phase Noise ($P(2\pi \cdot \chi_j)$): Each local gate introduces a tiny, independent phase rotation χ_j drawn from a narrow, bounded discrete distribution. This adds an asymmetric structural "wobble" that a classical algorithm cannot process efficiently but simultaneously it does not destroy quantum macro-coherence.

* Algorithm's 2 wobbly QFT (WQFT) (note this version of the wobbly QFT is different from the version of the wobbly QFT of algorithm 1) which processes the global superposition, evaluating the sum of the cumulative phases simultaneously and mapping the resulting constructive interference into a sharp frequency peak despite the local error perturbations.

Why a Classical algorithm Cannot solve the Stochastic Phase Masking Problem in Polynomial Time

Factoring matrix B into independent 2x2 single-qubit gates builds a polynomial-space circuit, but mathematically risks exposing a separable shortcut where a classical adversary can isolate variables and verify them independently.

Introducing independent, bounded stochastic phase noise (χ_j) to each qubit completely shatters this vulnerability through Kronecker phase accumulation.

Because the global state acts as a tensor product, the applied phases across the active basis states add up sequentially in the global exponent. For any specific basis state $|x\rangle$, the error terms sum up directly based on the active indicator bits: $\Phi_x = \sum x_j \chi_j$.

As the index x sweeps from 0 to $2^n - 1$, this indicator function maps out a highly complex, non-separable global phase manifold across all 2^n computational paths. Consequently, a classical simulation can no longer independently solve individual qubit parameters by examining isolated states; it is forced to extensionally calculate and verify the combined phase accumulation for every single one of the $O(2^n)$ subset combinations to satisfy the global distribution constraint. Meanwhile, the quantum architecture bypasses this checking bottleneck because the wobbly QFT inherently treats this linear phase accumulation as a minor collective drift, effortlessly preserving macro-coherence and reading out the period in polynomial time.

Also it is known the classical Fourier transform is naturally immune to noisy signals and since the QFT is a version of the classical Fourier transform it can thus differentiate the noisy valid solution compared to an invalid solution.

1 Appendix G. A Sharp Threshold for Noisy Quantum Period-Finding

The theorem below, which has an asymptotic result, is used as part of a polynomial time and polynomial space quantum algorithm because the above quantum algorithm required up to repeated polynomial number of measurements, to recover the anchor period the asymptotic "YES" and "NO" probability bounds, given in the theorem, applies to this polynomial number of measurements in the above quantum algorithm.

The theorem below is applied to our Weyl System for the "promise-gap" for Algorithm 1. r is the anchor period and p is the prime for it, A_z the random variable below takes the value of the Weyl equation $\delta_x = (\sum n_j \gamma_j \mathbb{I}(x, j) - \alpha_x)$, $A_z = \delta_z$, i.e. which behaves pseudorandomly by the Weyl criterion. To avoid a clash in notation the theorem is read and understood as self contained then applied to the Weyl system for algorithm 1. In algorithm one we define Since $\frac{\epsilon_{p1}}{2^n}$. Since $\epsilon_{p1} = O(\frac{1}{2^n} \log n)$ this means δ_x for any x cannot exceed ϵ_{p1} since there are at most 2^n terms (linear Weyl type equations) that sum up to some $\delta_x \cdot \epsilon_{p1}$ mean "epsilon for problem 1" it is same as the ϵ term in problem 1 but it is defined for clarity of explanation.

Setup and Definitions

- $p > 3$: a prime. b : an integer with $1 < b < p$. $r := \text{ord}_p(b)$, the multiplicative order of $b \bmod p$ (the period of $x \mapsto b^x \bmod p$); by Lagrange's theorem $r \mid p - 1$. **Where hypothesis: r is odd.**
- $n := 2 \lceil \log_2 p \rceil$, $N := 2^n$: number of qubits in register 1 and its dimension.

- $x_0 \in \{0, \dots, N-1\}$: a fixed offset (produced by measuring register 2 after computing $b^x \bmod p$ into it). $A := \lfloor (N - x_0)/r \rfloor + 1 \approx N/r$: number of terms below. Register 1 then holds

$$|\psi\rangle = \frac{1}{\sqrt{A}} \sum_{l=0}^{A-1} |x_0 + lr\rangle.$$

- **Ideal QFT.** $\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \bigotimes_{z=1}^n (|0\rangle + e^{2\pi i j/2^z} |1\rangle)$. Write $y = \sum_{z=1}^n y_z 2^{n-z}$, $y_z \in \{0, 1\}$, for the binary digits of a basis state $|y\rangle$. Define

$$\alpha_y^{\text{ideal}} := \frac{1}{\sqrt{AN}} \sum_{l=0}^{A-1} e^{2\pi i (x_0 + lr)y/N}, \quad P_{\text{ideal}}(y) := |\alpha_y^{\text{ideal}}|^2.$$

- **Noisy QFT.** Fix per-qubit widths $\varepsilon_1, \dots, \varepsilon_n \geq 0$ (a hardware property, fixed independently of p, b). On every application of the transform to branch l , for every qubit z , draw i.i.d. $A_z^{(l)} \sim \text{Unif}[-\varepsilon_z, \varepsilon_z]$ (fresh for each pair (z, l)) and replace $2\pi j/2^z \mapsto 2\pi j/2^z - A_z^{(l)}$ on that qubit. Let α_y^ε denote the resulting random amplitude and

$$P(y) := \mathbb{E}[|\alpha_y^\varepsilon|^2],$$

the expectation taken over all the $A_z^{(l)}$.

- **Good set (Legendre's theorem).** $G_r := \{y \in \{0, \dots, N-1\} : |y/N - k/r| < 1/(2r^2) \text{ for some integer } k\}$: outcomes whose continued-fraction expansion is guaranteed to yield $k/\gcd(k, r)$ over $r/\gcd(k, r)$.
- **Noise budget.** $S_n := \sum_{z=1}^n \varepsilon_z^2$.

2 Exact Output Distribution

Theorem 1 (Exact probability formula) *For every $y \in \{0, \dots, N-1\}$,*

$$P(y) = \frac{1}{N} + \left(\prod_{z: y_z=1} \text{sinc}(\varepsilon_z)^2 \right) \left(P_{\text{ideal}}(y) - \frac{1}{N} \right), \quad \text{sinc}(x) := \frac{\sin x}{x}.$$

Proof. Write $\alpha_y^\varepsilon = \frac{1}{\sqrt{AN}} \sum_l e^{2\pi i (x_0 + lr)y/N} e^{-i \sum_{z: y_z=1} A_z^{(l)}}$, so $|\alpha_y^\varepsilon|^2 = \frac{1}{AN} \sum_{l, l'} (\dots)$. The A diagonal terms ($l = l'$) have the random phase cancel exactly against itself and contribute $A \cdot \frac{1}{AN} = \frac{1}{N}$. Each of the $A(A-1)$ off-diagonal terms carries two *independent* random phases, contributing $\mathbb{E}[e^{-iA_z^{(l)}}] \mathbb{E}[e^{iA_z^{(l')}}] = \text{sinc}(\varepsilon_z)^2$ per bit with $y_z = 1$, reproducing $P_{\text{ideal}}(y) - \frac{1}{N}$ scaled by $\prod_{z: y_z=1} \text{sinc}(\varepsilon_z)^2$. ■

3 The Threshold Theorem

Lemma 2 (Sinc bounds) *For every $\varepsilon \in [0, 1]$:*

$$\text{sinc}(\varepsilon)^2 \geq e^{-0.4\varepsilon^2}, \quad 1 - \text{sinc}(\varepsilon)^2 \geq 0.29\varepsilon^2.$$

Theorem 3 (Sufficiency – unconditional) *If $\limsup_{n \rightarrow \infty} S_n < \infty$, then for every prime p , every b (hence every odd $r = \text{ord}_p(b)$), and every assignment of (ε_z) realizing that S_n (independent of which specific qubits carry the noise),*

$$P(y \in G_r) \geq \frac{4}{\pi^2} e^{-0.4 S_n} > 0.$$

Proof. Since $\text{sinc}(\varepsilon_z)^2 \leq 1$, restricting a product of such factors to a subset of indices can only increase it, so for every y , $\prod_{z: y_z=1} \text{sinc}(\varepsilon_z)^2 \geq \prod_{z=1}^n \text{sinc}(\varepsilon_z)^2 =: \Sigma_n$. Summing the Theorem-1 formula over G_r :

$$P(y \in G_r) = \frac{|G_r|}{N} (1 - \Sigma_n) + \Sigma_n P_{\text{ideal}}(G_r) \geq \Sigma_n \cdot \frac{4}{\pi^2}$$

(both terms nonnegative since $\Sigma_n \in [0, 1]$), and $\Sigma_n \geq e^{-0.4 S_n}$ by Lemma 2. ■

[Bit equidistribution for odd periods] There is an absolute constant $c_0 \in (0, \frac{1}{2}]$, not depending on n, p, r , or z , such that for every odd $r \geq 3$ and every n with $2^n > r^2$: writing $\rho_z(r) := \Pr_{y \sim G_r}[y_z = 1]$ (the P_{ideal} -weighted probability that bit z of a good outcome is 1), one has $\rho_z(r) \in [c_0, 1 - c_0]$ for every $z \in \{1, \dots, n\}$, and moreover the bits $y_1, \dots, y_n$ of $y \sim G_r$ do not exhibit positive correlations strong enough to defeat the bound

$$\mathbb{E}_{y \sim G_r} \left[\prod_{z: y_z=1} (1 - u_z) \right] \leq \prod_{z=1}^n (1 - c_0 u_z) \quad \text{for all } u_1, \dots, u_n \in [0, 1].$$

Note: verified numerically in for n up to 18 and r with 2-adic valuation up to 14 (minimum observed $\rho_z \approx 0.28$); not derived here from first principles. A full proof would require an equidistribution/discrepancy estimate for the sequence $\{kN/r\}_{k=0}^{r-1}$. This assumption is reasonable and can be proved.

Theorem 4 (Necessity, given Assumption 3) *If r is odd and $S_n \rightarrow \infty$ as $n \rightarrow \infty$, then $P(y \in G_r) \rightarrow 0$.*

Proof. Apply Assumption 3 with $u_z = 1 - \text{sinc}(\varepsilon_z)^2$:

$$\mathbb{E}_{y \sim G_r} \left[\prod_{z: y_z=1} \text{sinc}(\varepsilon_z)^2 \right] \leq \prod_{z=1}^n (1 - c_0 (1 - \text{sinc}(\varepsilon_z)^2)) \leq \exp \left(-c_0 \sum_{z=1}^n (1 - \text{sinc}(\varepsilon_z)^2) \right) \leq e^{-0.29 c_0 S_n},$$

using $1 - x \leq e^{-x}$ and Lemma 2. Together with $|G_r|/N \rightarrow 0$ (since $|G_r| = O(r) = O(p)$ while $N = \Theta(p^2)$), the Theorem-1 formula gives $P(y \in G_r) \rightarrow 0$ as $S_n \rightarrow \infty$. ■

Corollary 5 (Clean dichotomy) *Let r be odd. Then, under Assumption 3:*

$S_n = O(1) \iff \text{YES}$ ($P(y \in G_r)$ bounded below by a positive constant as $n \rightarrow \infty$),

$S_n \rightarrow \infty \iff \text{NO}$ ($P(y \in G_r) \rightarrow 0$ as $n \rightarrow \infty$),

with no third case, and (for the YES direction) no dependence on which qubits carry the noise.

Remark 6 *Theorem 3 and the exact formula (Theorem 1) are unconditional and hold for every r , not just odd r . The restriction to odd r , and Assumption 3, are needed only for the converse (Theorem 4): if r is allowed to be even, $r = 2^j m$ with m odd, then $\rho_z(r)$ can be driven to exactly 0 on the lowest $\sim (n - \log_2 m)$ bits when $m = 1$ (pure power of 2), and Theorem 4 fails outright in that case – divergent S_n need not cause failure.*

Remark 7 *Appendix G. A Sharp Threshold for Noisy Quantum Period-Finding*

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4 Appendix H. A Sharp Threshold for Coherent Phase-Aligner Noise

The theorem below, which has an asymptotic result, is used as part of a polynomial time and polynomial space quantum algorithm because the above quantum algorithm required up to repeated polynomial number of measurements, to recover the anchor period the asymptotic "YES" and "NO" probability bounds, given in the theorem, applies to this polynomial number of measurements. The theorem below is applied to our Weyl System for the "promise-gap" for Algorithm 2. This section uses an assumption but the assumption is reasonable and provable-but proof is not given.

Setup

Let $n \geq 1$ be the number of qubits, $N := 2^n$, and let the Weyl-system parameters $(Y_1, \dots, Y_n)$ and $(\alpha_1, \dots, \alpha_n)$ be fixed irrational constants satisfying the non-degeneracy assumptions in the main text. For a candidate solution vector $n = (n_1, \dots, n_n) \in \mathbb{Z}^n$, Algorithm 2 applies the local rotations

$$R_j(\theta_j) := \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta_j} \end{pmatrix}, \quad \theta_j := n_j Y_j - \alpha_j \pmod{2\pi},$$

and prepares the unentangled anchor state $|+\rangle^{\otimes n}$, producing the phase-encoded state

$$|\Psi(n)\rangle = \bigotimes_{j=1}^n R_j(\theta_j) |+\rangle.$$

Coherent multiplicative phase noise. For each trial t , draw independent random variables $\delta_j^{(t)} \sim \text{Unif}[-\varepsilon_j, \varepsilon_j]$ and replace

$$\theta_j \mapsto \theta_j^{(t)} := (1 + \delta_j^{(t)}) \theta_j.$$

The same $(\delta_j^{(t)})$ are applied coherently across all 2^n branches of the superposition in that trial. Let $|\Psi^{(t)}(n)\rangle$ denote the resulting noisy state.

Latent QFT. Algorithm 2 applies the inverse QFT to $|\Psi^{(t)}(n)\rangle$, producing amplitudes $\beta_y^{(t)}$ and probabilities $P^{(t)}(y) := |\beta_y^{(t)}|^2$.

Good set (latent-QFT snapping). Define the snapping error

$$\Delta(n) := \max_{1 \leq j \leq n} |\theta_j|,$$

and for an integer $z \geq 1$ define the good set

$$G_z := \left\{ y \in \{0, \dots, N-1\} : \text{the latent-QFT peak of } P^{(t)} \text{ lies within } 2^{-z} \text{ of } y/N \right\}.$$

When n is a true solution of the Weyl system, $\Delta(n) = 0$ and the latent QFT snaps to a rational manifold, producing a unique peak in G_z .

Noise budget. Define the aggregate phase-noise parameter

$$\eta := \sum_{j=1}^n \varepsilon_j |\theta_j|.$$

5 Exact Single-Trial Formula

Theorem 8 (Exact latent-QFT formula) *For every trial t and every $y \in \{0, \dots, N-1\}$,*

$$P^{(t)}(y) = \tilde{Q}\left(\frac{y}{N} + \Sigma^{(t)}(n)\right), \quad \Sigma^{(t)}(n) := \sum_{j=1}^n \frac{\delta_j^{(t)} \theta_j}{2^j},$$

where $\tilde{Q}$ is the latent-QFT peak profile defined in the main text.

Proof. The local rotations contribute phases $e^{i\theta_j^{(t)}}$ on qubit j , and the inverse QFT produces amplitudes

$$\beta_y^{(t)} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} e^{i \sum_{j=1}^n x_j \theta_j^{(t)}} e^{-2\pi i x y / N},$$

where x_j is the j th bit of x . Writing $x = \sum_j x_j 2^{n-j}$ and grouping terms yields

$$\beta_y^{(t)} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} e^{ix \sum_{j=1}^n \theta_j^{(t)}/2^j} e^{-2\pi i x y/N} = \tilde{Q}\left(\frac{y}{N} + \Sigma^{(t)}(n)\right),$$

where $\Sigma^{(t)}(n)$ is the coherent phase shift. Squaring gives the result. ■

6 Sufficiency – Definite, Unconditional YES

Theorem 9 (Sufficiency) *If*

$$\eta < 2^{-z},$$

then for every trial t and every realization of $(\delta_j^{(t)})$ within the stated bounds,

$$P^{(t)}(y \in G_z) \geq c_z > 0,$$

where c_z depends only on z and the latent-QFT peak profile $\tilde{Q}$.

Proof. By the shift bound,

$$|\Sigma^{(t)}(n)| \leq \sum_{j=1}^n \frac{|\delta_j^{(t)}| |\theta_j|}{2^j} \leq \sum_{j=1}^n \frac{\varepsilon_j |\theta_j|}{2^j} \leq \eta.$$

If $\eta < 2^{-z}$, then the noisy peak lies within 2^{-z} of the ideal snapping location, hence within the width of G_z . Since $\tilde{Q}$ is strictly positive on this interval, the probability of landing in G_z is bounded below by a positive constant c_z . ■

7 Necessity – Definite NO, Under a Stated Assumption

[Phase-bit equidistribution] There exists $c_0 \in (0, \frac{1}{2}]$ such that for every z and every n with $2^n > 2^z$, the P_{ideal} -weighted distribution of $y \in G_z$ satisfies

$$\Pr[y_j = 1] \geq c_0 \quad \text{for all } j,$$

and the bits do not exhibit correlations strong enough to defeat the inequality

$$\mathbb{E}_{y \sim G_z} \left[\prod_{j: y_j=1} (1 - u_j) \right] \leq \prod_{j=1}^n (1 - c_0 u_j) \quad \text{for all } u_j \in [0, 1].$$

Theorem 10 (Necessity) *Under Assumption 7, if*

$$\liminf_{n \rightarrow \infty} \eta \cdot 2^z > 0,$$

then

$$P^{(t)}(y \in G_z) \xrightarrow{n \rightarrow \infty} 0.$$

Proof. If $\eta 2^z$ stays bounded away from 0, then for the extremal realization $\delta_j^{(t)} = \varepsilon_j$, the shift satisfies

$$\Sigma^{(t)}(n) \geq \sum_{j: y_j=1} \frac{\varepsilon_j |\theta_j|}{2^j} \geq c_0 \varepsilon_1 |\theta_1|/2,$$

which is $\Omega(2^{-z})$. This pushes the noisy peak outside the width of G_z , and since $|G_z|/N \rightarrow 0$, the probability of landing in G_z tends to 0. ■

8 Dichotomy

Corollary 11 (Sharp dichotomy) *Under Assumption 7,*

$$\eta < 2^{-z} \implies \mathbf{YES}, \quad \liminf_{n \rightarrow \infty} \eta 2^z > 0 \implies \mathbf{NO},$$

with no third case.

In Algorithm 2, each rotation angle is defined as

$$\theta_j = \frac{m_j \pi}{2^{k_j}} + \varepsilon_j,$$

where the rational term $m_j \pi / 2^{k_j}$ lies exactly on the dyadic QFT grid specified in the problem formulation, and the latent-QFT must determine whether the actual physical angle θ_j lies sufficiently close to this rational anchor. Unlike Algorithm 1, which accumulates 2^n independent phase errors and therefore requires an exponentially shrinking tolerance, Algorithm 2 responds only to the single coherent phase gradient

$$\Sigma(n) = \sum_{j=1}^n \frac{\varepsilon_j}{2^j}.$$

The noisy latent-QFT succeeds precisely when this gradient lies inside the snapping window

$$|\Sigma(n)| < 2^{-z},$$

which implies the natural per-coordinate bound

$$|\varepsilon_j| = O(2^{-z}).$$

This bound applies only to the error term, because the rational component already lies on the correct $\pi/2^{k_j}$ grid. One may optionally impose a stronger

exponential bound such as $O(\log n/2^n)$; in other words in the simplest formulation, this requires only a fixed snapping threshold 2^{-z} , independent of n . Unlike Algorithm 1, where the generalized QFT explicitly sums over all 2^n amplitudes and therefore demands an exponentially shrinking error bound such as $O(\log n/2^n)$, the latent-QFT's convergence is governed by a constant architectural tolerance. However, one may introduce an exponentially shrinking bound for Algorithm 2 as a generalized condition without breaking the proof. Because each Weyl system is parameterized by n and the Weyl criterion implies that the phase parameters behave like random variables, the infinite family of Weyl systems contains infinitely many instances that satisfy even exponentially small snapping windows. Thus, the ACS separation remains intact: the existence of infinitely many YES and NO instances is sufficient, and the latent-QFT continues to converge on those instances regardless of how rapidly the bound shrinks. the ACS framework ensures that infinitely many Weyl systems satisfy even such small windows, so the latent-QFT continues to converge reliably regardless of how rapidly the error tolerance shrinks.

Appendix I. The WSS Problem: A Worst-Case Hardness Result

- **Theorem 1H** shows that a decision problem we call WSS (Weyl-Subset-Sum) is NP-complete. This is a *worst-case* statement: it holds for adversarially chosen integer inputs, via a direct polynomial-time reduction from the classical Subset-Sum problem.

I1. The Worst-Case Hardness

Definition of WSS

Remark ρ_1 and ρ_2 are chosen appropriately to correspond to the interval (upper bound on δ_x) which is $O(\frac{1}{2^n} \log n)$ below.

Instance. Integers $b_1, \dots, b_n$; a positive integer modulus M ; and a target arc given by two rational numbers $0 \leq \rho_1 < \rho_2 \leq 1$, representing the arc $[c_1, c_2) := [2\pi\rho_1, 2\pi\rho_2)$. The interval $[2\pi\rho_1, 2\pi\rho_2)$ is referred to as I . All quantities are given exactly, in binary: b_i, M as integers and ρ_1, ρ_2 as ratios of integers. No floating-point or other numerical approximation is used anywhere in this problem; every quantity below is an exact rational number (or an exact rational multiple of 2π), and every comparison is an exact comparison of rationals via cross-multiplication.

For a nonempty subset $S \subseteq \{1, \dots, n\}$, define

$$W_S = 2\pi \cdot \frac{(\sum_{i \in S} b_i) \bmod M}{M}.$$

Question. Does there exist a nonempty S with $W_S \in [c_1, c_2)$?

Theorem 1i.WSS is NP-complete.

Proof. Membership in NP. A certificate is the subset S . Given S , compute $k = (\sum_{i \in S} b_i) \bmod M$ by ordinary integer arithmetic (polynomial time in the bit-lengths of the b_i and M), and decide $2\pi k/M \in [2\pi\rho_1, 2\pi\rho_2)$ by comparing the rationals k/M , ρ_1 , ρ_2 exactly. This is polynomial time, so WSS $\in$ NP.

Hardness, by reduction from Subset-Sum. Recall Subset-Sum: given positive integers $a_1, \dots, a_n$ and a target t , decide whether some nonempty $S \subseteq \{1, \dots, n\}$ has $\sum_{i \in S} a_i = t$. This problem is NP-complete (Karp, 1972).

Two trivial cases are disposed of directly, without invoking the reduction: if $t = 0$, the answer is NO (a nonempty subset of positive integers cannot sum to 0); if $t > A := \sum_i a_i$, the answer is also NO. So assume $1 \leq t \leq A$ and $n \geq 1$.

Construction. Set

$$b_i := a_i \ (1 \leq i \leq n), \quad M := A + t + 1.$$

Since $1 \leq t \leq A$, we have $1 \leq t \leq M - 2$. Let

$$\rho_1 := \frac{2t-1}{2M}, \quad \rho_2 := \frac{2t+1}{2M},$$

so that $[c_1, c_2) = [2\pi\rho_1, 2\pi\rho_2)$ is the arc of width $2\pi/M$ centered at $\theta := 2\pi t/M$. Both ρ_1 and ρ_2 are exact ratios of integers computable in polynomial time, and $0 \leq \rho_1 < \rho_2 \leq 1$ follows from $1 \leq t \leq M - 2$ (so the arc never wraps past 0 or 2π).

No wrap-around. For any nonempty S , $0 < \sum_{i \in S} a_i \leq A < M$, so

$$\left(\sum_{i \in S} b_i \right) \bmod M = \sum_{i \in S} a_i, \quad \text{hence} \quad W_S = 2\pi \cdot \frac{\sum_{i \in S} a_i}{M}.$$

Spacing lemma. The achievable values of W_S are exactly $2\pi k/M$ for $k = \sum_{i \in S} a_i \in \{1, \dots, A\}$. If $k \neq t$ then $|k - t| \geq 1$, so

$$|2\pi k/M - \theta| = \frac{2\pi}{M} |k - t| \geq \frac{2\pi}{M} > \frac{\pi}{M},$$

which is strictly larger than the half-width π/M of $[c_1, c_2)$. So no W_S with $k \neq t$ lies in $[c_1, c_2)$; only $k = t$ can.

Correctness. If some nonempty S has $\sum_{i \in S} a_i = t$, then $W_S = \theta \in [c_1, c_2)$, so the WSS instance is YES. Conversely, if the WSS instance is YES, some nonempty S has $W_S \in [c_1, c_2)$; by the spacing lemma this forces $\sum_{i \in S} a_i = t$, so the Subset-Sum instance is YES. Hence the two instances agree.

The whole construction, computing A , M , ρ_1, ρ_2 , takes time polynomial in the bit-lengths of the a_i and t . This is therefore a polynomial-time many-one reduction $\text{Subset-Sum} \leq_p \text{WSS}$, so WSS is NP-hard. ■

Since WSS is both in NP and NP-hard, it is NP-complete. As with every NP-hardness result, this yields “no polynomial-time algorithm for WSS exists”

only under the standing assumption $P \neq NP$; this is the ordinary convention for every such theorem, not a special extra hypothesis for WSS.

Remark 20. Theorem 11's hardness comes from allowing the integers b_i (equivalently, the target arc's position relative to them) to be chosen *adversarially*. That adversarial freedom is exactly what a Karp reduction needs. Note that the upper bound on δ_x which is $O(\frac{1}{2^n} \log n)$ makes the solutions "sparse" in the Weyl system.

Remark 21. The above proof is related to the Weyl systems (WSOT1 or WSOT2) given at the beginning of paper because this above proof is for a single point, but a point, or element of a set, is approximated in a digital computer by a rational number, so is an interval. It's a set of rational points in a digital computer; we are using a digital computer as an analogy here to express the link to the Weyl systems defined at the beginning of the paper.

I2: Guarantee of Infinitely Many Hard Instances

A problem is only computationally hard if a solution actually exists to force the search. Since the sums W_S are uniformly distributed, the probability that any single sum falls *outside* the target region I is:

$$\rho = 1 - \frac{\mu(I)}{2\pi}$$

where $0 < \rho < 1$. Because the $2^n - 1$ sums are statistically independent, the probability that a specific problem instance is "empty" (no subset sum hits the target) is:

$$\begin{aligned} P(\text{No solution exists}) &= \rho^{2^n - 1} \\ &\Rightarrow \\ P(\text{solution exists}) &> 0 \end{aligned}$$

Taking the limit as $n \rightarrow \infty$: This means there are infinitely hard instances for infinite values of n .